

Small-molecule Activation of TFEB Alleviates Niemann-Pick Disease Type C via Promoting Lysosomal Exocytosis and Biogenesis

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eLife Assessment

This study reports that activation of TFEB promotes lysosomal exocytosis and clearance of cholesterol from lysosomes, the strength of evidence for which is **convincing** with appropriate and validated methodology in line with current state-of-the-art. The significance of the findings is **important** in the context of Niemann-Pick Disease Type C as well as other subfields.

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Abstract

Niemann–Pick disease type C (NPC) is a devastating lysosomal storage disease characterized by abnormal cholesterol accumulation in lysosomes. Currently, there is no treatment for NPC. Transcription factor EB (TFEB), a member of the microphthalmia transcription factors (MiTF), has emerged as a master regulator of lysosomal function and promoted the clearance of substrates stored in cells. However, it is not known whether TFEB plays a role in cholesterol clearance in NPC disease. Here, we show that transgenic overexpression of TFEB, but not TFE3 (another member of MiTF family) facilitates cholesterol clearance in various NPC1 cell models. Pharmacological activation of TFEB by sulforaphane (SFN), a previously identified natural small-molecule TFEB agonist by us, can dramatically ameliorate cholesterol accumulation in human and mouse NPC1 cell models. In NPC1 cells, SFN induces TFEB nuclear translocation via a ROS-Ca²⁺-calcineurin dependent but MTOR-independent pathway and upregulates the expression of TFEB-downstream genes, promoting lysosomal exocytosis and biogenesis. While genetic inhibition of TFEB abolishes the cholesterol clearance and exocytosis effect by SFN. In the NPC1 mouse model, SFN dephosphorylates/activates TFEB in brain and exhibits potent efficacy of rescuing the loss of Purkinje cells and body weight. Hence, pharmacological upregulating lysosome machinery via targeting TFEB represents a promising approach to treat NPC and related lysosomal storage diseases, and provides the possibility of TFEB agonists ie SFN as potential NPC therapeutic candidates.

Introduction

Lysosomes are essential organelles for the degradation and recycle of damaged complex substrates and organelles (Xu and Ren 2015 [\[1\]](#)). In recent years, lysosomes have emerging roles in plasma membrane repair, external environmental sensing, autophagic cargo sensing, and proinflammatory response, thereby regulating fundamental processes such as cellular clearance and autophagy (Tsunemi, Perez-Rosello et al. 2019 [\[2\]](#)). Mutations in genes encoding lysosomal proteins could result in more than approximately 70 different lysosomal storage disorders (Fraldi, Klein et al. 2016 [\[3\]](#)). Niemann-Pick disease type C (NPC) is a rare lysosomal storage disorder caused by mutation in either *NPC1* or *NPC2* gene. Deficiency in *NPC1* or *NPC2* protein results in late endosomal/lysosomal accumulation of unesterified cholesterol (Sarkar, Carroll et al. 2013 [\[4\]](#), Spampinato, Feeney et al. 2013 [\[5\]](#)). Clinical symptoms of NPC include hepatosplenomegaly, progressive neurodegeneration and central nervous system dysfunction ie. seizure, motor impairment, and decline of intellectual function (Carstea, Morris et al. 1997 [\[6\]](#)). So far there is no FDA-approved specific therapy for NPC, although miglustat, approved to treat type I Gaucher disease, has been used for NPC treatment in countries including China, Canada and European Union (Pineda, Wraith et al. 2009 [\[7\]](#), Pineda, Perez-Poyato et al. 2010, Wraith, Vecchio et al. 2010 [\[8\]](#), Chien, Peng et al. 2013 [\[9\]](#))

Transcription factor EB (TFEB), a member of the microphthalmia/TFE transcription factor (MITF) family, is identified as a master regulator of lysosome and autophagy by controlling the “coordinated lysosomal expression and regulation” (CLEAR) network, covering genes associated to lysosomal exocytosis and biogenesis, and autophagy (Sardiello and Ballabio 2009 [\[10\]](#), Martini-Stoica, Xu et al. 2016 [\[11\]](#), Napolitano and Ballabio 2016 [\[12\]](#)). In normal condition, TFEB is phosphorylated by mTOR kinase and kept in cytosol inactively (Martina and Rosa 2018 [\[13\]](#), Napolitano, Esposito et al. 2018). Under stress conditions ie. starvation or oxidative stress, TFEB is dephosphorylated and actively translocates into nucleus, promoting the expression of genes associated with lysosome and autophagy (Medina, Diego et al., Zhang, Cheng et al. 2016 [\[14\]](#), Puertollano, Ferguson et al. 2018 [\[15\]](#)). TFEB overexpression or activation results in increased number of lysosomes, autophagy flux and exocytosis (Medina, Fraldi et al. 2011 [\[16\]](#), Settembre, Di Malta et al. 2011 [\[17\]](#), Giatromanolaki, Kalamida et al. 2015 [\[18\]](#), Xu, Du et al. 2020 [\[19\]](#)), which may trigger cellular clearance. In fact, it is reported that TFEB overexpression promotes cellular clearance and ameliorates the phenotypes in a variety of neurodegenerative diseases including Alzheimer’s, Parkinson’s, and Huntington’s. (Sardiello, Palmieri et al. 2009 [\[20\]](#), Tsunemi and La Spada 2012 [\[21\]](#), Decressac and Björklund 2013 [\[22\]](#), Polito, Li et al. 2014 [\[23\]](#), Ballabio 2016 [\[24\]](#), Napolitano and Ballabio 2016 [\[25\]](#)). Moreover, upregulation of TFEB has been reported to benefit lysosomal storage diseases (LSDs), such as Pompe disease (Argüello, Balboa et al. 2021 [\[26\]](#)). However, it is not known whether upregulation of TFEB by genetic or pharmacological methods is sufficient to increase lysosomal function and alleviate NPC phenotypes *in vitro* or *in vivo*. If so, TFEB may be a putative target for NPC treatment and manipulating lysosomal function via small-molecule TFEB agonists may have broad therapeutic potential for NPC.

Results

Transgenic overexpression or pharmacological activation of TFEB reduces cholesterol accumulation in various human NPC1 cell models

TFEB and TFE3 are identified as key transcription factors regulating lysosome and autophagy biogenesis (Sardiello, Palmieri et al. 2009 [\[1\]](#)). To investigate the role of TFEB/TFE3 in cellular cholesterol levels in NPC1 cells, HeLa cells were treated with U18666A (2.5 μ M, an inhibitor of the endosomal/lysosomal cholesterol transporter NPC1), a drug that has been widely used to induce NPC phenotype in cell models (Poh, Shui et al. 2012 [\[2\]](#)) (Hereafter HeLa NPC1 cells represent U18666A-treated HeLa cells). Cellular cholesterol levels were measured by the well-known cholesterol-marker filipin (Lu, Liang et al. 2015 [\[3\]](#)). As shown in **Fig. 1A, B**, in [\[4\]](#) HeLa NPC1 cells overexpressing TFEB-mCherry or TFEB^{S211A}-mCherry (S211 non-phosphorylatable mutant, a constitutively active TFEB) (Wang, Gao et al. 2015 [\[5\]](#)), the cellular cholesterol levels (filipin) were significantly diminished compared to non-expressed or mCherry-only transfected NPC1 cells. In contrast, TFE3-GFP overexpression displayed no obvious reduction of cholesterol levels in HeLa NPC1 cells (**Fig. S1A, B** [\[4\]](#)). Hence, TFEB but not TFE3 contributes to cholesterol reduction in NPC1 cells.

Pharmacological activation of TFEB is an emerging strategy for LSD treatment (Levine and Kroemer 2008 [\[6\]](#)). We have previously identified a natural TFEB agonist-sulforaphane (SFN), which is also an activator of cellular antioxidant NFE2L2/Nrf2 pathway, dramatically mitigating oxidative stress commonly associated with metabolic and age-related diseases including NPC diseases (Corssac, Campos-Carraro et al. 2018 [\[7\]](#), Li, Shao et al. 2021 [\[8\]](#), Liu, Fu et al. 2021 [\[9\]](#)). Based on our previous data that SFN (10-15 μ M, 3-24 h) potently activates TFEB in various cell lines, promoting lysosomal function (Li, Shao et al. 2021 [\[8\]](#)), thus, in this study, we set SFN treatment dose to 15 μ M and treatment time to 4-24 h according to individual experiment. We hypothesize that SFN may contribute to cholesterol clearance in NPC disease. To evaluate this hypothesis, we first examined the effect of SFN on cellular cholesterol levels in HeLa NPC1 cell model using filipin staining. As shown in **Fig. 1C, D** [\[4\]](#), U18666A treatment significantly increased filipin signals in bright perinuclear granules, which were well co-localized with lysosomal marker LAMP1, but not with ER marker calnexin in HeLa cells (**Fig. S2A, B** [\[4\]](#)), suggesting that U18666A-induced cholesterol accumulation presents in the late endosome/lysosome compartment, but not in ER. Moreover, when HeLa NPC1 cells were further challenged with SFN (15 μ M, 12-24 h), a dramatically reduction of cholesterol accumulation (filipin intensity) by more than 30% was observed in lysosomes (**Fig. 1C-E** [\[4\]](#), **Fig. S2C, D** [\[4\]](#)). These results confirmed that U18666A interferes with the egress of free cholesterol from endosomes/lysosomes as previously reported by others (Lange and Steck 1994 [\[10\]](#), Davis, Shin et al. 2021 [\[11\]](#)) and SFN reduces lysosomal cholesterol in NPC1 cells.

In our previous study, we identified that SFN activates TFEB via a ROS-Ca²⁺-calcineurin-mediated but MTOR (mechanistic target of rapamycin kinase)-independent mechanism (Li, Shao et al. 2021 [\[8\]](#)). Thus, we further validated this mechanism of SFN in HeLa NPC1 cell model by performing immunofluorescence experiments. In HeLa NPC1 cells, SFN (15 μ M, 24 h) induced robust TFEB translocation from the cytosol to the nucleus (**Fig. 1F, G** [\[4\]](#)). SFN treatment (15 μ M, 24 h) also resulted an increase in the mRNA levels of TFEB downstream genes, including genes required for autophagy (*ULK1*, *SQSTM1* and *ATG5*) and lysosome biogenesis-*LAMP1*, using quantitative real-time PCR (Q-PCR) (**Fig. 1H** [\[4\]](#)). Consistent with previous finding, BAPTA-AM (a membrane-permeable Ca²⁺ chelator), FK506 and cyclosporin A (CsA) (calcineurin inhibitors) and N-acetylcysteine (NAC, a ROS scavenger) can robustly block SFN-induced TFEB nuclear translocation in HeLa NPC1 cells (**Fig. S3** [\[4\]](#)), suggesting that in NPC1 cells SFN induces TFEB nuclear translocation via a ROS-Ca²⁺-calcineurin dependent mechanism. A key mechanism of TFEB activation is Ca²⁺-dependent dephosphorylation of TFEB by protein phosphatases (Medina, Diego et al., Martina and Puertollano 2018 [\[12\]](#)). We next investigated the specific phosphorylated site on TFEB by SFN in NPC1 cells. Following 24 h exposure to SFN, S211 and S142-TFEB phosphorylation

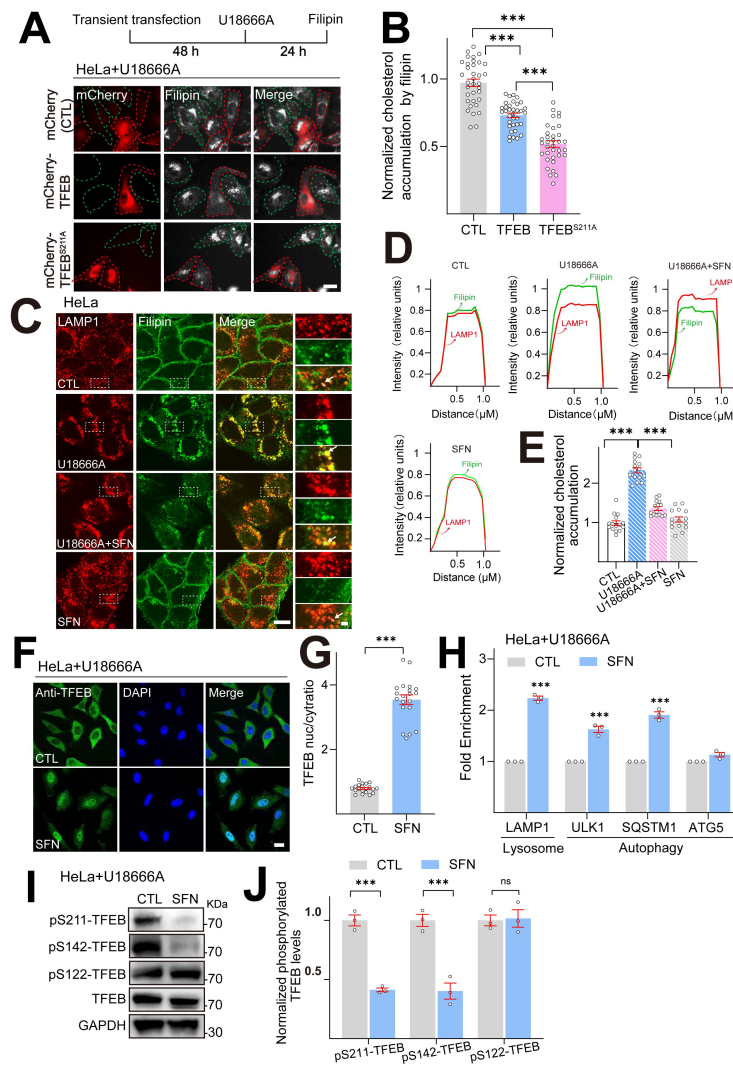


Figure 1.

TFEB overexpression or pharmacological activation of TFEB ameliorates cholesterol accumulation in U18666A-induced HeLa NPC1 model.

(A) Overexpression of TFEB/TFEB^{S211A} reduced cellular cholesterol levels in U18666A-induced HeLa NPC1 model. Filipin staining of HeLa transiently transfected by mCherry, mCherry-TFEB and mCherry-TFEB^{S211A} plasmid for 48 h, followed by U18666A (2.5 μM) for 24 h. Overlay phase contrast images are shown together with the red (mCherry-TFEB/TFEB^{S211A}) and white (filipin). In each image the red circles point to the successful transfected cells, green circles represent the untransfected cells. Scale bar, 20 μm. **(B)** Quantification of cholesterol accumulation from A. N=30 randomly selected cells from n=3 independent experiments. **(C)** SFN reduces lysosomal (LAMP1) cholesterol accumulation (Filipin) in HeLa NPC1 cells. HeLa cells were exposed to U18666A (2.5 μM) in the absence or the presence of SFN (15 μM) for 24 h. Each panel shows fluorescence images taken by confocal microscopes. The red signal is LAMP1-mCherry driven by stable transfection, and the green signal is Filipin. **(D)** Each panel shows the fluorescence intensity of a line scan (white line on the blown-up image) through the double labeled object indicated by the white arrow. Scale bar, 20 μm or 2 μm (for zoom-in images). **(E)** Quantification of cholesterol levels shown in C. N=15 randomly selected cells from n=3 experiments. **(F)** SFN (15 μM, 24 h) induced TFEB nuclear translocation in HeLa NPC1 cells. Nuclei were counterstained with DAPI (blue). Scale bar, 20 μm. **(G)** Average ratios of nuclear vs. cytosolic TFEB immunoreactivity shown in F. N=20 randomly selected cells from n=3 experiments. **(H)** SFN induced mRNA expression of TFEB target genes in HeLa NPC1 model. HeLa cells were cotreated with U18666A (2.5 μM) and SFN (15 μM) for 24 h (n=3). **(I)** Western blot analysis of TFEB phosphorylation by SFN (15 μM, 24 h) in HeLa NPC1 cells. **(J)** Quantification of ratios of pS211-, pS142-, pS122-TFEB vs. total TFEB as shown in I. (n=3). For all the panels, average data are presented as mean ± s.e.m.; *** P < 0.001.

were significantly decreased (**Fig. 1I, J**), while S122 phosphorylation was not affected. These results indicate that TFEB is dephosphorylated at S211 and S142 residues by SFN in HeLa NPC1 model. TFEB nuclear shuffling is regulated by the activity of MTOR, which is regulated by its phosphorylation status (Martina, Chen et al. 2012). SFN reportedly activates TFEB in a mTOR-independent manner (Li, Shao et al. 2021). Consistently, no significant inhibition of phosphorylated MTOR (p-MTOR) or RPS6KB1 (p-RPS6KB1) can be observed in HeLa NPC1 cells treated with SFN (15 μ M, 24 h) (**Fig. S4A, B**). Therefore, in HeLa NPC1 cells SFN-induced TFEB activation is unlikely to be mediated by MTOR inhibition. We also examined the possibility of the direct effect of SFN on NPC1 by western blotting. As shown in **Fig. S5**, SFN (15 μ M, 24 h) treatment did not affect NPC1 expression in human NPC1-patient fibroblasts. Give the well-known role of SFN as a NFE2/Nrf2 inducer, we also validated whether SFN can induce Nrf2 activation in NPC cells. As shown in **Fig. S6A, B**, SFN (15 μ M, 24 h) induced robust Nrf2 nuclear translocation from the cytosol to the nucleus in HeLa NPC1 and NPC primary mouse embryonic fibroblasts (MEF) cells.

Next, we verified the effect of SFN in another NPC1 cell model by knockdown (KD) *NPC1* with specific siRNA (Liao, Wang et al. 2015; Hoglinger, Burgoyne et al. 2019). In *NPC1* KD HeLa cells with more than 80 % knockdown efficiency (**Fig. 2A**), SFN can induce TFEB nuclear translocation (**Fig. 2B, C**) and upregulate its downstream gene expression (**Fig. 2D**). Notably, SFN (15 μ M, 24 h) treatment significantly reduced cholesterol levels in *NPC1* KD cells (**Fig. 2E, F**). Similar results were observed in human NPC1-patient fibroblast cells. SFN treatment robustly promoted TFEB nuclear translocation (**Fig. 2G, H**) and cholesterol clearance (**Fig. 2I, J**). Taken together, these experiments demonstrate that pharmacological activation of TFEB activation by SFN could promote cellular cholesterol clearance in various NPC *in vitro* models.

TFEB is required in SFN-promoted cholesterol clearance in NPC1 cells

We next investigated whether TFEB is required for SFN-promoted cholesterol clearance using two strategies. On one hand, HeLa cells were treated with the siRNAs specifically against TFEB or transiently overexpressed with mCherry-TFEB. The efficiency of siRNA KD and overexpression was evaluated by western blot (**Fig. 3A**). In the si*TFEB*-transfected HeLa NPC1 cells, SFN-promoted (15 μ M, 24 h) cholesterol clearance was almost abolished (**Fig. 3B, C**). In contrast, in scrambled siRNA-transfected NPC1 cells, the cholesterol levels were significantly reduced by more than 30% by SFN. Notably, TFEB overexpression-induced reduction of cholesterol can be further boosted about 20% by SFN treatment (**Fig. 3B, C**). On the other hand, HeLa *TFEB* KO cells was constructed by the CRISPR/Cas9 tool (**Fig. 3D**). Consistently, SFN failed to diminish cholesterol in HeLa *TFEB* KO cells, whereas re-expression of a recombinant TFEB restored the cholesterol clearance by SFN (**Fig. 3E, F**). Collectively, these data indicate that TFEB is specifically required for lysosomal cholesterol reduction upon SFN treatment.

SFN promotes lysosomal exocytosis and biogenesis in human NPC1 models in a TFEB-dependent manner

We then studied the mechanism by which SFN led to diminish cellular cholesterol in NPC1 cells. Considering the established role of TFEB activation in lysosomal exocytosis, we verified the hypothesis that SFN-induced cholesterol clearance through upregulation of lysosomal exocytosis using surface LAMP1 immunostaining (**Fig. 4A**). When lysosomal exocytosis processes, luminal lysosomal membrane proteins can be detected on the extracellular side of the plasma membrane (PM) by measuring surface expression of LAMP1 (lysosomal associated membrane protein 1), a resident marker protein of late endosomes and lysosomes (referred to as “lysosomes” for simplicity hereafter) with a monoclonal antibody against a luminal epitope of LAMP1 (Reddy, Caler et al. 2001). After incubation with SFN (15 μ M) for 24 h, HeLa NPC1 cells exhibited a

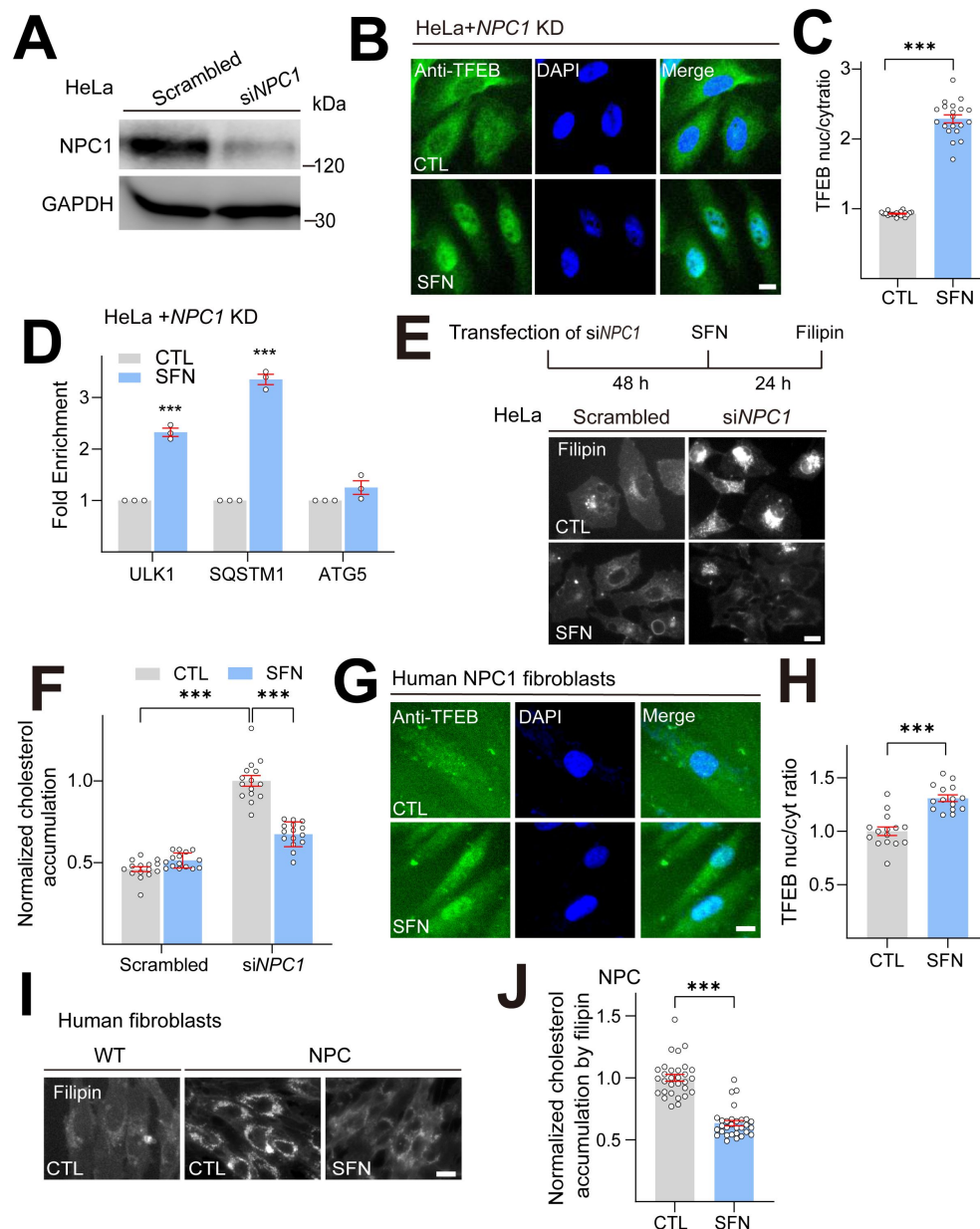


Figure 2.

SFN promotes cholesterol clearance in various human NPC1 cell models.

(A) Western blot analysis of the KD efficiency of a specific NPC1-targeting siRNA in HeLa cells ($n = 3$ independent repeats). **(B)** SFN induced TFEB nuclear translocation in NPC1 KD HeLa cells. Detection of TFEB immunoreactivity in HeLa cells transiently transfected with siNPC1 for 48 h, followed by SFN (15 μ M) treatment for 24 h. Scale bar, 20 μ m. **(C)** Average ratios of nuclear vs. cytosolic TFEB immunoreactivity shown in **B**. $N=20$ randomly selected cells from 3 independent repeats. **(D)** In NPC1 KD HeLa cells, SFN (15 μ M, 24 h) upregulated expression of TFEB target genes ($n=3$ independent repeats). **(E)** SFN promoted cholesterol clearance in NPC1 KD HeLa cells. HeLa cells were transiently transfected by siNPC1 for 48 h, followed by SFN (15 μ M) treatment for 24 h. Scale bar, 20 μ m. **(F)** Quantification of cholesterol accumulation in NPC1 KD HeLa cells shown in **E**. $N=15$ randomly selected cells from 3 independent repeats. **(G)** SFN (15 μ M, 24 h)-induced TFEB nuclear translocation in human NPC1 fibroblasts. Nuclei were counterstained with DAPI (blue). Scale bar, 20 μ m. **(H)** Quantification of nuclear vs. cytosolic TFEB ratio as shown in **G**. $N=20$ randomly selected cells from at least 3 independent experiments. **(I)** SFN promoted cholesterol clearance in human NPC1-patient fibroblasts. Human NPC1 fibroblasts were treated with SFN (15 μ M, 24 h) and Filipin staining was carried out. Scale bar, 20 μ m. **(J)** Quantification of cholesterol accumulation as shown in **I**. $N=30$ randomly selected cells from 3 independent repeats. For all the panels, average data are presented as mean $\pm$ s.e.m.; *** $P < 0.001$.

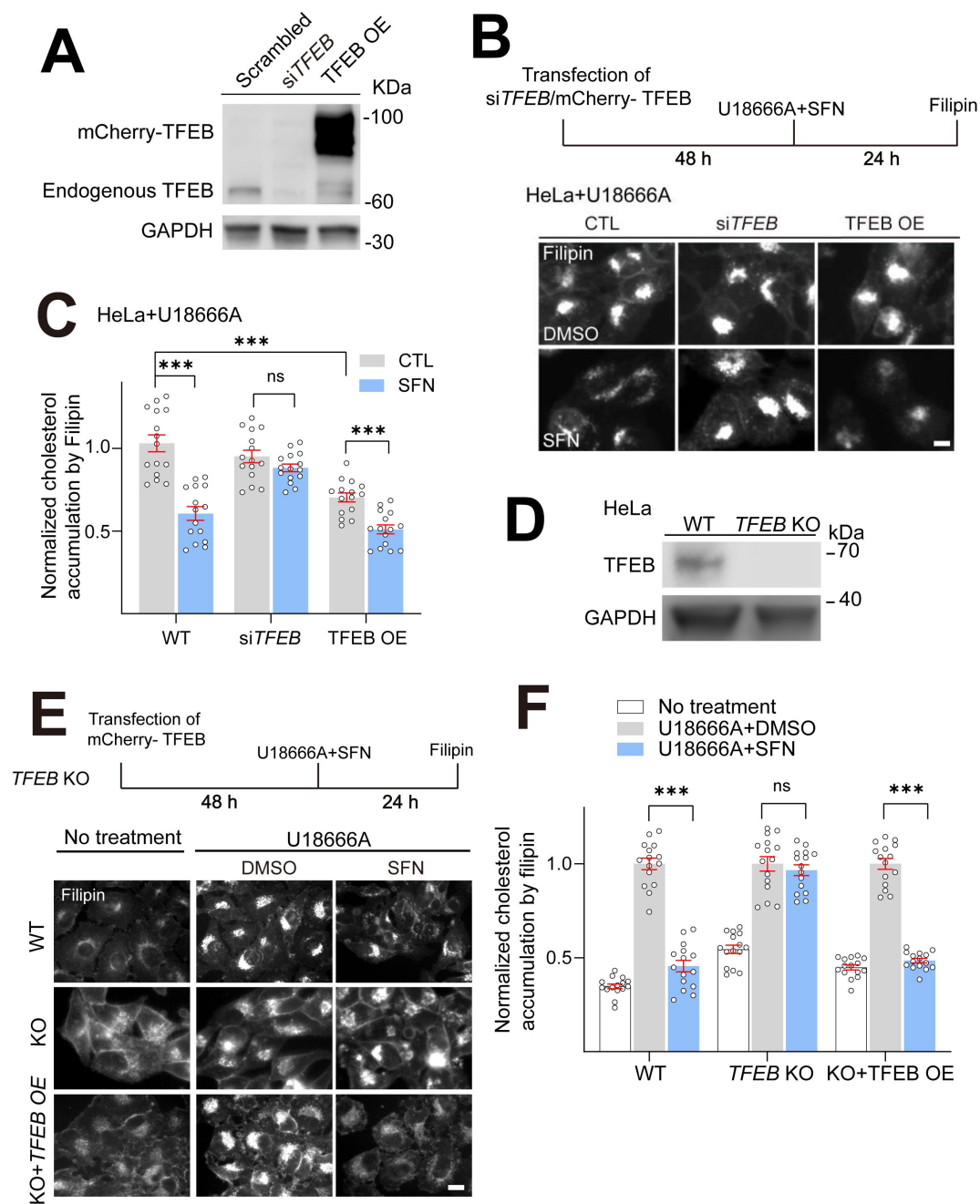


Figure 3.

TFEB is required for SFN-promoted cholesterol clearance.

(A) Western blot analysis of the efficiency of siTFEB KD and mCherry-TFEB OE in HeLa cells. **(B)** HeLa cells were transfected with siTFEB or mCherry-TFEB for 48 h, followed by cotreatment with U18666A (2.5 μ M) and SFN (15 μ M) for 24 h and cholesterol accumulation was analyzed by Filipin assay. Scale bar, 20 μ m. **(C)** Quantification of cholesterol levels as shown in **B**. N=15 randomly selected cells from 3 independent repeats. **(D)** Western blot analysis the efficiency of TFEB KO in HeLa cells. **(E)** HeLa, HeLa TFEB KO and HeLa TFEB KO cells transiently expressing mCherry-TFEB (TFEB OE, 48 h) were cotreatment with U18666A (2.5 μ M) and SFN (15 μ M) for 24 h and cholesterol levels were analyzed by Filipin assay. Scale bar, 20 μ m. **(F)** Quantification analysis of cholesterol accumulation as shown in **E**. N=15 randomly selected cells from at least 3 independent experiments. Average data are presented as mean $\pm$ s.e.m.; *** P < 0.001.

dramatic increase in the signal of LAMP1 staining in the PM (Surface LAMP1 signal colocalized with DiO, a PM marker) compared to untreated control cells (**Fig. 4A, B**), suggesting an increase of lysosomal exocytic process. Likewise, SFN promoted lysosomal exocytosis in primary macrophage cells (**Fig. S7A**). In contrast, this increase of surface LAMP1 signal by SFN was significantly reduced in *TFEB* KO cells (**Fig. S7B**), demonstrating that SFN-induced lysosomal exocytosis is *TFEB*-dependent.

A direct consequence of lysosomal exocytosis is the release of lysosomal contents into the cell culture medium (Rodriguez, Webster et al. 1997). We then quantified the release of free cholesterol/cholesteryl ester into the medium upon SFN treatment in HeLa NPC1 cells using the Cholesterol/Cholesteryl Ester assay Kit. This assay allows to detect total cholesterol or free cholesterol in the presence or absence of cholesterol esterase; the levels of cholesteryl ester are determined by subtracting the levels of free cholesterol from total cholesterol. As shown in **Fig. 4C**, the levels of free cholesterol released into the medium treated were significantly increased with SFN (15 μ M, 24 h) treatment compared to untreated control, suggesting that SFN can stimulate the release of free cholesterol into medium. Notably, the difference of released cholesteryl ester with or without SFN treatment has not been observed and the percentage of cholesteryl ester in total cholesterol released to medium were less than 10% under all conditions tested. Meanwhile, we further examined the effect of SFN on the release of lysosomal enzymes- β -hexosaminidase (NAGase) and acid phosphatase (ACP). In HeLa NPC1 cells treated with SFN (15 μ M, 24 h), significantly higher levels of lysosomal enzymes (NAGases and ACP) were detected in the medium compared with untreated controls, but not in *TFEB* KO cells (**Fig. 4D**). Taken together, these results indicate that SFN induces an active movement of lysosomes toward the PM and increases lysosomal exocytosis in a *TFEB*-dependent manner.

Previously we reported that SFN could increase lysosome biogenesis and regulate lysosomal function, which contribute to ROS reduction in NPC models (Li, Shao et al. 2020). We then studied whether SFN activate lysosomal machinery in NPC1 cells. In HeLa NPC1 cells, SFN (15 μ M, 12 h) treatment significantly increased the immunofluorescence intensity of LAMP1 (**Fig. 4E, F**). Likewise, similar results were observed in NPC1 KD cells (**Fig. S8A, B**). Lysosomal enzymes operate better under acidic conditions, and the degradation-active lysosomes can be tracked using LysoTracker, a fluorescent acidotropic probe (Li, Gu et al. 2019). We observed significant increases of LysoTracker staining in HeLa NPC1 cells following 12 h treatment with SFN (15 μ M) (**Fig. 4G, H**) as well as in NPC1 KD cells (**Fig. S8C, D**), yet LAMP1 staining was also increased (**Fig. 4E, F**; **Fig. S8A, B**). Thus lysosomal pH was more accurately determined using a ratiometric dye-pHrodo Green Dextran. When the fluorescence ratios (pHrodo Green Dextran /CF555) were calibrated to pH values, we found that SFN treatment induced significant lysosomal hyperacidity (**Fig. 4I, J**). Collectively, these results suggest that SFN promotes lysosome function and biogenesis in human NPC1 model cells, consistent with our previous report.

Notably, we then examined the *TFEB* expression in two NPC cell lines (HeLa NPC1 cells and NPC1-patient fibroblast cells). As shown in **Fig. S9A, B**, *TFEB* expression levels in both NPC cell models were significantly decreased compared to WT, indicating that *TFEB* expression may be inhibited in NPC cells. Interestingly, the basal levels of lysosome biogenesis in NPC cells were comparable with WT cells (**Fig. 4D-G**, **Fig. S8A-D**), suggestive of compensatory changes caused by *TFEB* downregulation. Taken together, excessive activation of *TFEB* in NPC cells can be targeted for cholesterol clearance via upregulation of lysosomal function and biogenesis.

SFN alleviates cholesterol accumulation

in primary NPC1^{-/-} MEF cells

To address the possible relevance of the SFN/*TFEB* axis in NPC pathology of mice experiments, we then investigated whether SFN is sufficient to reduce cholesterol and regulates lysosomal function via *TFEB* activation in primary murine cells. Primary mouse embryonic fibroblasts (MEF) were

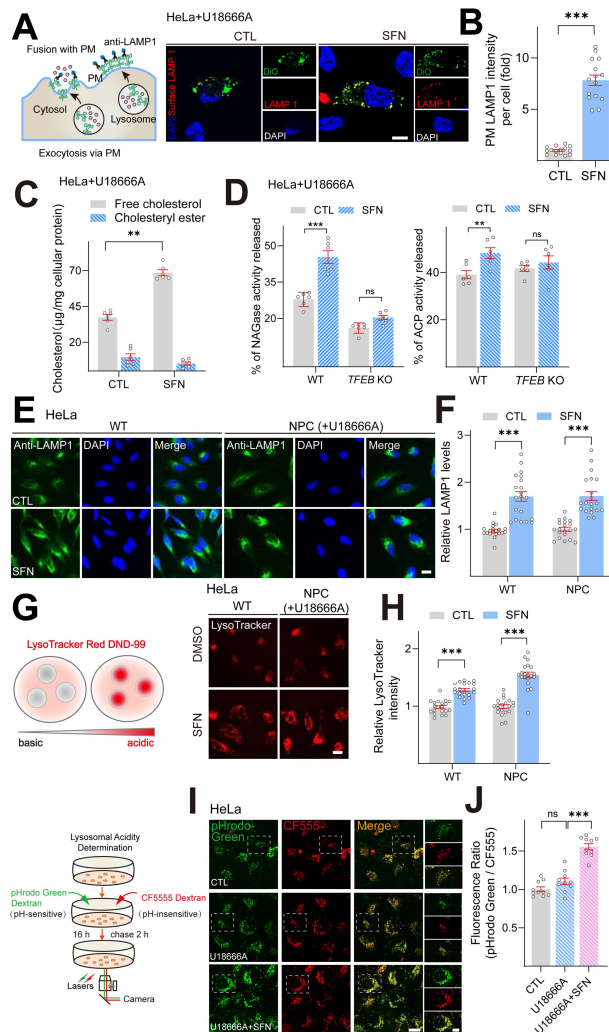


Figure 4.

SFN promotes lysosomal exocytosis and biogenesis in NPC1 cell models.

(A) Confocal microscopy images showing the exposure of LAMP1 on the PM in nonpermeabilized HeLa NPC1 cells treated with SFN (15 μM) for 24 h using an antibody against LAMP1 luminal portion. Nuclei were counterstained with DAPI (blue). Scale bar, 20 μm. **(B)** Quantitative analysis of LAMP1 levels on the PM in HeLa NPC1 cells shown in **A**. Bars represent the fold increase of LAMP1 fluorescence on PM in SFN treated cells. N=15 randomly selected cells from 3 independent repeats. **(C)** SFN increased the release of free cholesterol into medium. HeLa cells were cotreated with U18666A (2.5 μM) and SFN (15 μM) for 24 h, and then examined for cholesterol. The levels of cholesterol in the medium or cell lysates were measured by cholesterol assay in a reaction mixture with (measuring total cholesterol content) or without (measuring free cholesterol content) cholesterol esterase enzyme (n=6 independent repeats). **(D)** SFN increased the release of lysosomal enzyme NAGases and ACP in HeLa NPC1 cells. HeLa cells were cotreated with U18666A (2.5 μM) and SFN for 24 h, and the activities of NAGases and ACP were analyzed in the medium and cell lysates (n=6 independent repeats). **(E)** LAMP1 staining in HeLa cells upon U18666A treatment (2.5 μM) in the presence and absence of SFN (15 μM). Nuclei were counterstained with DAPI (blue). Scale bar, 20 μm. **(F)** Quantification analysis of LAMP1 immunofluorescence shown in **E**. N=20 randomly selected cells from at least 3 independent experiments. **(G)** Effects of SFN on lysosome acidity. HeLa cells were treated with 2.5 μM U18666A (24 h) in the presence and absence of 15 μM SFN (12 h) and lysosomal pH was analyzed by LysoTracker Red DND-99 (50 nM). Scale bar, 20 μm. **(H)** Quantification of LysoTracker intensity shown in **G**. N=20 randomly selected cells from at least 3 independent experiments. **(I)** Effects of SFN on lysosomal acidity using a ratiometric pH dye. HeLa cells were treated with U18666A (2.5 μM) in the presence and absence of SFN (15 μM), lysosomal pH was determined using a ratiometric pH dye combination (pHrodo Green dextran and CF555 dextran). Scale bar, 20 μm or 2 μm (for zoom-in images). **(J)** Quantification analysis of lysosomal pH shown in **I**. Randomly selected cells from at least 3 independent experiments. For all the panels, data are presented as mean ± s.e.m.; ***P* < 0.01, ****P* < 0.001.

freshly prepared from *NPC1*^{-/-} mice (from Jackson's laboratory). SFN (15 μ M) treatment for 24 h dramatically increased TFEB nuclei signal in the *NPC1*^{-/-} MEF cells (**Fig. 5A, B**). Next, we analyzed the effect of SFN on cholesterol clearance in *NPC1*^{-/-} MEF cells. SFN (15 μ M) treatment for 72 h exhibited substantial cholesterol reduction, whereas SFN treatment for 24 h showed relative weaker cholesterol clearance in MEF cells (**Fig. 5C, D**) compared with human NPC1 cells (**Fig. 1C**), suggesting that human NPC cells are more sensitive to SFN treatment compared to mouse NPC cells. We further tested whether SFN promotes lysosomal biogenesis and function in *NPC1*^{-/-} MEF cells. Following 24 h treatment with SFN (15 μ M), a significant increase of LAMP1 staining (**Fig. 5E, F**) and LysoTracker intensity (**Fig. 5G, H**) were observed in *NPC1*^{-/-} MEF cells. Collectively, these results suggest that SFN regulates TFEB-mediated lysosomal function axis and promotes cellular cholesterol clearance in NPC MEF cells.

SFN alleviates the loss of Purkinje cells and body weight in *Npc1*^{-/-} mice

Considering that SFN promotes lysosomal cholesterol clearance in both human and murine NPC1 cell models (**Fig. 1C, 2C, 5C**) and reportedly penetrates blood–brain barrier (BBB) (Kim, Kim et al. 2012, Mao, Yang et al. 2018, Tavakkoli, Iranshahi et al. 2019). We next investigated that whether SFN targets/activates TFEB in brain, 4-week-old BALB/cJ mice were intraperitoneally injected with SFN (50 mg/kg) or vehicle for 12 h and brain tissues including cerebellum and hippocampus were collected and pS211-TFEB/TFEB levels were measured by western blotting. As shown in **Fig. 6A, B**, we observed a significant decrease of pS211-TFEB protein in brain tissues with SFN treatment compared to vehicle, suggesting that TFEB in the brain was directly targeted by SFN treatment. This is the first time that SFN was shown to directly active TFEB in the brain. We then evaluated the *in vivo* therapeutic efficacy of SFN, *NPC1*^{-/-} mice (4-week-old) were treated with SFN (30 or 50 mg/kg) by daily intraperitoneal injection for 4 weeks. Purkinje cells located in the cerebellum are the most susceptible to NPC1 loss and exhibit significant selective loss in the anterior part of the cerebellum (Sarna, Larouche et al. 2003). Purkinje cells in cerebellum sections were stained by Calbindin and quantified by recording the number of surviving cells in lobules/mm of Purkinje cell layer. As shown in **Fig. 6C, D**, little survival of Purkinje cells in vehicle-treated *NPC1*^{-/-} cerebellum, in contrast, daily injection of SFN (50 mg/kg) in NPC1 mice prevented a degree of Purkinje cell loss particularly in the lobule IV/V of cerebellum. Body weight is another important indicator of therapeutic efficacy in NPC mice. Typically, NPC1 mice weight plateaus at 6–7 weeks of age, and then progressively declines. Notably, we observed that daily i.p. injection of SFN (50 mg/kg) exhibited a significantly improvement in weight loss of NPC1 mice (**Fig. 6E**). However, SFN treatment has no effect on liver and spleen enlargement of NPC1 mice (data not shown). Collectively, our results demonstrated that pharmacological activation of TFEB by small-molecule agonist can mitigate NPC neuropathological symptoms *in vivo*.

Discussion

We have demonstrated in the current study that genetic overexpression of TFEB, but not TFE3 can dramatically mitigate cholesterol accumulation in NPC cells. Pharmacological activation of TFEB by SFN, a previously identified TFEB agonist (Li, Shao et al. 2021), significantly promoted cholesterol clearance in human and mouse NPC cells, while genetic inhibition (KO) of TFEB blocked SFN-induced cholesterol clearance. This clearance effect exerted by SFN was associated with upregulated lysosomal exocytosis and biogenesis. Notably, SFN is reportedly BBB-permeable, assuring a good candidate for efficient delivery to the brain, which is essential for targeting neurodegenerative phenotypes in neurological diseases including NPC. In the NPC mouse models, SFN exhibits *in vivo* efficacy of suppressing the loss of Purkinje cells and maintaining body weight.

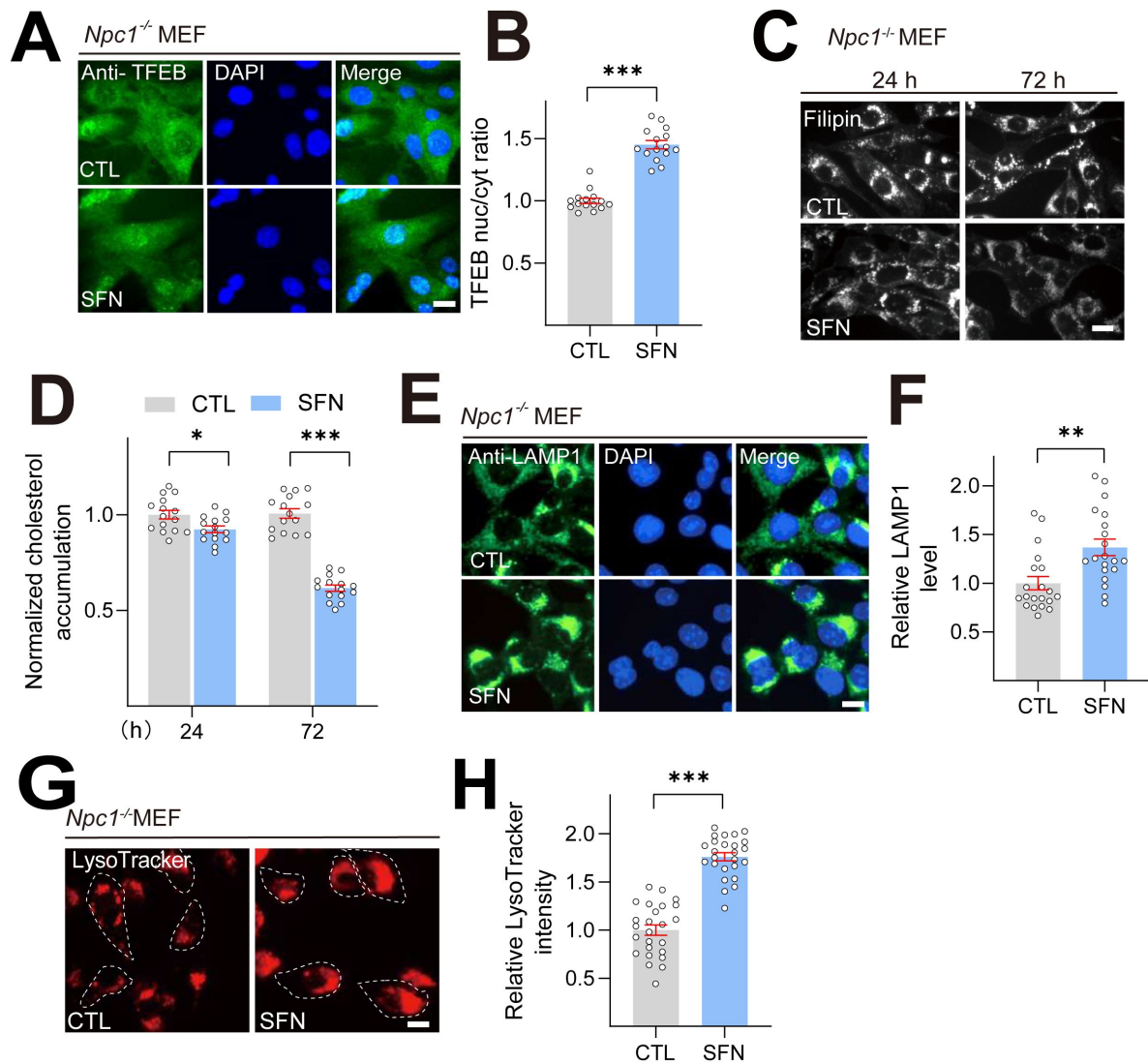


Figure 5.

SFN ameliorates cholesterol accumulation in *Npc1*^{-/-} MEF cells.

(A) SFN (15 μ M) treatment induced TFEB nuclear translocation in NPC1 MEF cells. Nuclei were counterstained with DAPI (blue). Scale bar, 20 μ m. (B) Average ratios of nuclear vs. cytosolic TFEB immunoreactivity shown in A. N=20 from 3 independent repeats. (C) SFN (15 μ M, 24-72 h) reduced cholesterol accumulation in NPC1 MEF cells by filipin assay. Scale bar, 20 μ m. (D) Quantification analysis of cholesterol accumulation levels shown in C. N=15 randomly selected cells from at least 3 independent experiments. (E) Effects of SFN (15 μ M, 12 h) on intensity of LAMP1 in NPC1 MEF cells. Scale bar, 20 μ m. (F) Quantification of LAMP1 intensity shown in E. N=20 randomly selected cells from at least 3 independent experiments. (G) Effects of SFN (15 μ M, 12 h) on lysosome acidity in MEF cells. Scale bar, 20 μ m. (H) Quantification analysis of LysoTracker intensity shown in G. N=20 randomly selected cells from at least 3 independent experiments. For all the panels, data are presented as mean $\pm$ s.e.m.; * P < 0.05, ** P < 0.01, *** P < 0.001.

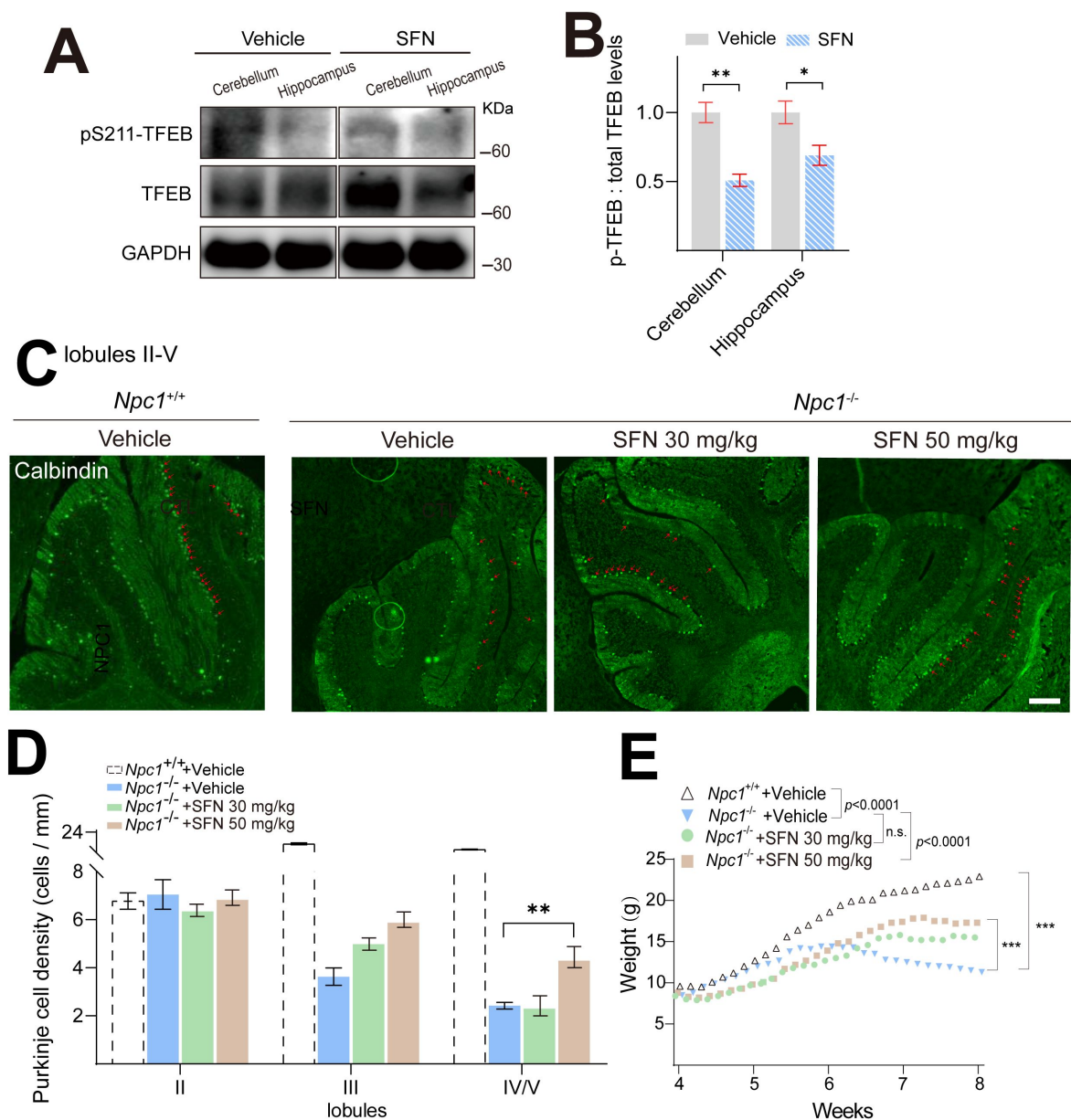


Figure 6.

SFN rescues the loss of Purkinje cells and body weight in NPC *in vivo* model mice.

(A) SFN promoted TFEB dephosphorylation in mice brain. 4-week-old BALB/cj mice were intraperitoneally (i.p) injected with SFN (50 mg/kg) or vehicle for 12 h and brain tissues including cerebellum and hippocampus were collected and subjected to detect pS211-TFEB and total TFEB levels by Western blotting. (B) Quantification of the ratios of p-TFEB vs. total TFEB as shown in A. (C) Cerebella from vehicle and SFN treated NPC mice were analyzed at 8 weeks of age for calbindin by immunohistochemistry. SFN and vehicle were intraperitoneally injected daily in 4-week-old NPC mice for 4 weeks. Scale bar = 200 μ m. (n=6 for each group). (D) Quantification of the number of Purkinje cells as indicated in the anterior lobules (II-V) as shown in C. (E) Body Weight was registered during the treatment. For all the panels, data are presented as mean $\pm$ s.e.m.; ** $P < 0.01$, *** $P < 0.001$.

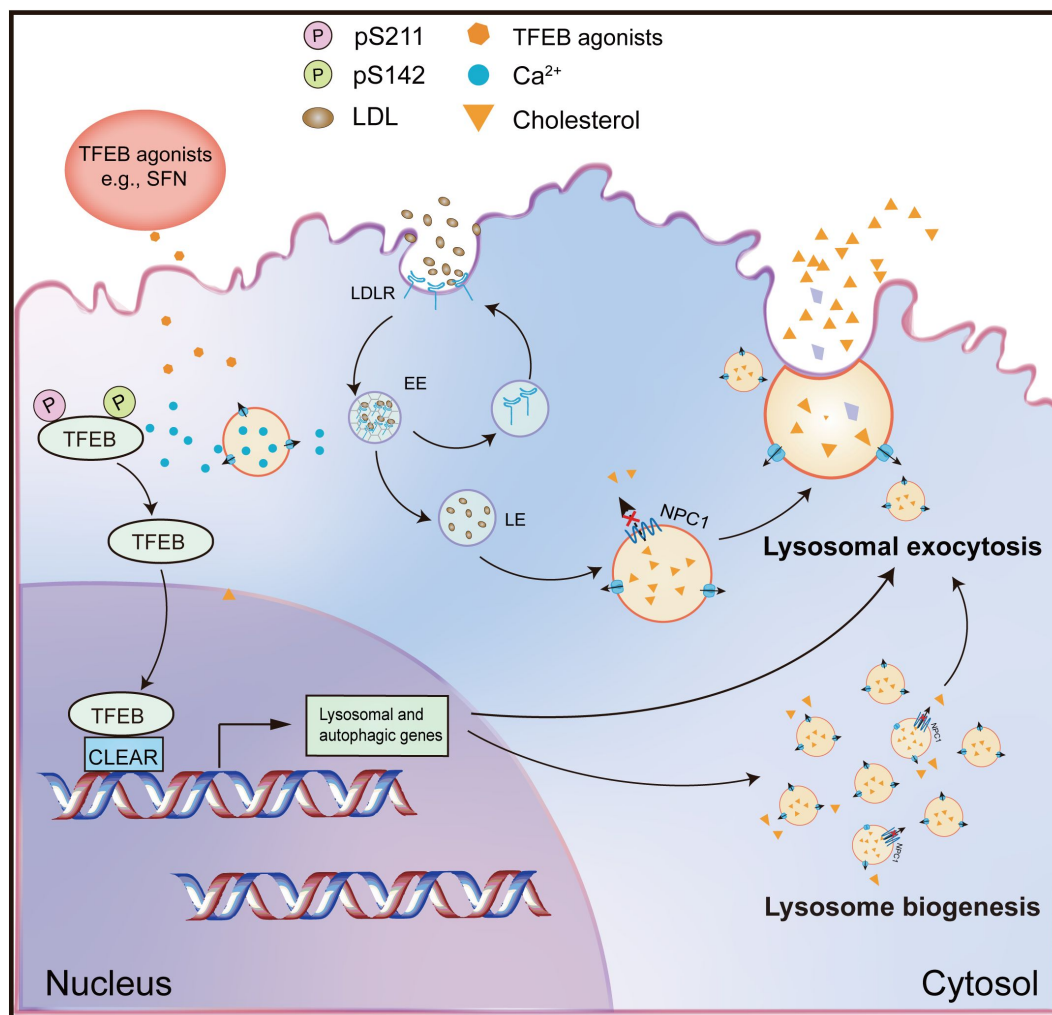


Figure 7.

A working scheme to illustrate that small-molecule TFEB agonist promotes cholesterol clearance in NPC1 model via TFEB-upregulated lysosomal exocytosis and biogenesis.

Pharmacological or genetic activation/overexpression of TFEB dramatically ameliorates cholesterol accumulation in NPC1 cells. Small-molecule, BBB-permeable TFEB agonist-SFN induces TFEB nuclear translocation by dephosphorylation of TFEB at S142 and S211 residues, promoting lysosomal biogenesis and exocytosis, resulting in mitigating lysosomal cholesterol levels.

Hence, genetically or pharmacologically targeting TFEB may represent a promising approach to treat NPC and manipulating lysosome function with small-molecule TFEB agonists may have broad therapeutic potentials.

The MiT/TFE family contains four factors: MITF, TFEB, TFE3, and TFEC, which share an identical basic region for DNA binding, and highly similar HLH and Zip regions for dimerization (Haq and Fisher 2011 [↗](#)). Many of the mechanistic insights into MiT regulation have been focused on TFEB and TFE3, which shares some overlapping functions. Surprisingly, in this study we found that overexpression of TFEB, but not TFE3 alleviated lysosomal cholesterol accumulation in NPC cells (Fig. 1A, B [↗](#), Fig. S1A, B [↗](#)). Moreover, studies reported that only TFEB overexpression, but not other MiT members upregulates lysosomal gene expression (Sardiello, Palmieri et al. 2009 [↗](#)). Thus, these results suggest that the functions exerted by TFEB and TFE3 in NPC may appear to be specialized.

As a proof of the role of TFEB activation in NPC, pharmacological activation of TFEB by SFN, a natural small-molecule TFEB agonist, promotes a dramatic lysosomal cholesterol-lowering effect in several genetic and pharmacological NPC cell models (Fig. 1C [↗](#), 5C [↗](#)). The concentration of SFN used in this study has no cytotoxicity towards the cell lines we used (Fig. S10 [↗](#)). SFN-induced lysosomal exocytosis and the increased population of lysosomes ready to fuse with the PM contribute to the cholesterol clearance by SFN (Fig. 4A-G [↗](#)). Previously we have shown that SFN can mitigate oxidative stress via a ROS-Ca²⁺-calcineurin-TFEB-mediated lysosomal function and autophagy flux (Li, Shao et al. 2021 [↗](#)). SFN, an electrophilic compound enriched in cruciferous vegetables such as broccoli, is a known potent inducer of NFE2L2/NRF2 in various cell types including NPC cells (Fig. S6A,B [↗](#)), a transcriptional factor that controls the expression of multiple detoxifying enzymes through antioxidant response elements (AREs) (Yamamoto, Kensler et al. 2018 [↗](#)). Notably, the promoter region of the *Nfe2l2* gene harbors a CLEAR site, and TFEB activation can upregulate the expression of *Nfe2l2* (Mansueto, Armani et al. 2017). Hence, TFEB, together with NFE2L2, function as key regulators of cellular redox. Oxidative stress is a major feature of NPC and has been attributed to neuronal damage, leading to the pathogenesis and progression of NPC (Vazquez, Balboa et al. 2012 [↗](#)). Elevated oxidative stress has been observed in the brain of NPC patient and mice (Smith, Wallom et al. 2009 [↗](#), Zampieri, Mellon et al. 2009 [↗](#)). NPC patients also show decreased antioxidant capacity (expressed as Trolox equivalents) and diminished activity of different antioxidant enzymes, which indicates a decrease in antioxidant defenses (Fu, Yanjanin et al. 2010 [↗](#)). Hence, the protective effect of SFN against NPC could attribute to a combination with antioxidant activity and cholesterol clearance via the lysosome-dependent, TFEB-mediated regulation. Therefore, pharmacological activation of TFEB may serve as a potential therapeutic strategy for NPC.

Materials and methods

Mammalian cell culture

HeLa cells were cultured in Dulbecco's Modified Eagle's Medium (DMEM; Thermo Fisher Scientific, 11195-065) supplemented with 10 % fetal bovine serum (FBS; Thermo Fisher Scientific, 10091148). NPC1 patient-derived fibroblast cells (clone GM03123) and a healthy control (clone GM03440) were obtained from the Coriell Institute for Medical Research (NJ, USA). Human fibroblast cells were maintained in modified Eagle's medium (Thermo Fisher Scientific, 1964643) supplemented with 15 % FBS, 2 mM glutamine (Thermo Fisher Scientific, 25030081) and 1 % penicillin-streptomycin. *NPC1*^{-/-} MEF cells were cultured in DMEM with 10 % FBS, 1 % penicillin-streptomycin and 1 % antibiotic-antimycotic (Thermo Fisher Scientific, 15240062). Macrophage cells were cultured in RPMI 1640 medium (Gibco, B122656) supplemented with 20 % FBS. Unless otherwise indicated, all cell cultures were maintained at 37 °C in a humidified 5 % CO₂ incubator.

Stable and CRISPR/Cas9 KO cell lines

GFP-TFEB stable HeLa cell line was kindly provided by Dr. Shawn Ferguson (Yale School of Medicine) (Zhang, Cheng et al. 2016 [DOI](#)). *TFEB* CRISPR-Cas9 KO HeLa cells were generated and characterized as reported previously (Li, Shao et al. 2020 [DOI](#)).

Plasmids/siRNA transfection

Plasmids including mCherry-TFEB, mCherry-TFEB^{S211A} and TFE3-GFP were maintained in our laboratory as previously described (Nezich, Wang et al. 2015 [DOI](#), Li, Shao et al. 2021 [DOI](#)). The siRNA sequences targeting human *TFEB* (5'-GAA AGG AGA CGA AGG UUC AAC AUC A-3') were purchased from Invitrogen. The siRNA sequences targeting human *NPC1* (5'-CAA UUG UGA UAG CAA UAU UTT-3') were chemically synthesized by GenePharma (Shanghai, China). HeLa cells were transfected with plasmids or siRNA using Lipofectamine 3000 reagent (Thermo Fisher Scientific, 2163785) or RNAi-Max reagent (Thermo Fisher Scientific, 13778150) in Opti-MEM (Thermo Fisher Scientific, 11058021), respectively. The efficiency of transfection was examined by Western blotting or Q-PCR.

Filipin staining

Cellular unesterified cholesterol were detected using a cell-based cholesterol assay kit (Abcam, ab133116). It provides a simple fluorometric method to detect the interaction of cholesterol and filipin III, which alters the filipin absorption and fluorescence spectra allowing visualization with a fluorescence microscope. Cells were cultured at 5×10^2 cells/well in black, clear-bottom 96-well plates and treated with compounds for the indicated conditions. After rinsing with PBS twice, cells were fixed with fixation solution for 10min followed by a PBS rinse twice. The cells were then stained with filipin III solution for 1 h at room temperature in the dark. The images were then captured by an Olympus IX73/Zeiss microscope. Image analysis was conducted using the image J software.

Immunofluorescence and confocal imaging

For immunofluorescence detection of TFEB and Nrf2, cells were grown on coverslips, fixed with 4 % PFA, permeabilized with 0.3 % Triton X-100 (Solarbio, T8200), followed by blocking in 1 % bovine serum albumin (BSA, Merck, B2064) in PBS for 1 h. Cells were then incubated with anti-TFEB (Cell Signaling Technology, 4240) or anti-Nrf2 antibody (Abcam, ab623521) at 4 °C overnight. After four washes with PBS, coverslips were incubated with secondary antibodies for 1 h and counterstained with DAPI for 10 min. For LAMP1 immunostaining, cells were fixed with 100 % methanol (−20 °C) for 10 min and then blocked with 1 % BSA in PBS for 1 h. The primary anti-LAMP1 (Abcam, ab24170) was used in this study. Finally, coverslips were mounted with Fluoromount-G (Southern Biotech, 0100-01) and ready for imaging.

Western Blotting

Cells were lysed with ice-cold RIPA buffer (Solarbio, R0010) in the presence of 1×protease inhibitor cocktail (Merck, P8340) and 1×phosphatase inhibitor cocktail (Abcam, GR304037-28) on ice for 20 min. Cells were then centrifuged and the supernatant was collected. The protein concentration of the supernatant was determined using BCA Protein Assay (Thermo Scientific, UA269551). Protein samples (20–40 µg) were then loaded and separated on SDS-polyacrylamide gradient gels (GenScript, M00654) followed by the transfer to polyvinylidene difluoride membranes (Merck, R7DA8778E). Western blot analysis was performed using primary antibodies against TFEB (1:500, Cell Signaling Technology, 4240 for cells, 1:1000, Bethyl Laboratories, A303-673A for mice tissue), pS122-TFEB (1:500, Cell Signaling Technology, 86843), pS142-TFEB (1:500, Millipore, 3321796), pS211-TFEB (1:500, Cell Signaling Technology, 37681), NPC1 (1:1000, Abcam, ab134113), LAMP1 (1:500, Abcam, ab24170), MTOR (Cell Signaling Technology, 2972), p-MTOR (Sigma-Aldrich,

SAB4504476), p-RPS6KB1/S6K1 (Cell Signaling Technology, 9234), RPS6KB1/S6K1 (Cell Signaling Technology, 2708), NPC1 (1:1000, Santacruz, sc271335) and GAPDH (1:10000, Sigma-Aldrich, G9545). Bound proteins were then detected with secondary antibodies against horseradish peroxidase-conjugated and enhanced chemiluminescence reagents (Thermo Fisher Scientific, 203-17071). The membranes were visualized by using an Li-COR Biosciences Odyssey Fc system and the band intensity was quantified using Image J software.

LAMP1 surface labeling

Cells were pretreated with SFN as indicated time. For the experiment of co-staining of DiO and surface LAMP1, cells were first incubated with 5 μ M DiO (HY-D0969, MCE) at 37°C for 30 min and washed with PBS twice to remove the unbound dye. Nonpermeabilized cells were then labeled with anti-human LAMP1 antibody (1:500, DSHB, H4A3), which recognizes a luminal epitope, at 4 °C for 1 h. Cells were then fixed in 2 % paraformaldehyde for 30 mins and incubated with Alexa Flour 488-conjugated secondary antibody at room temperature for 1 h. After PBS wash for three times, cells were counterstained with DAPI for 10 min and images were captured by Zeiss confocal microscope.

Lysosomal pH imaging

To measure lysosomal luminal pH, live cells were treated with chemicals as indicated condition, followed by incubation with 50 nM LysoTracker Red DND-99 (Thermo Scientific, L7528) for 15-30 min. Cells were then washed with PBS for three times. Images were captured by an Olympus/Zeiss microscope. Fluorescence intensities were quantified using the Image J software.

Lysosomal luminal acidity was also evaluated with the fluorescence ratio between a pH-sensitive dye pHrodo™ Green conjugated dextran-10 kd (P35368, Invitrogen) and a pH-insensitive dye CF555 conjugated dextran-10 kd (80112, Biotium). Briefly, cells were seeded on coverslips and loaded with 20 μ g/mL of each dextran overnight. Cells were then chased in medium without dye for 3 hours before imaging. The HEPES buffered DMEM medium without phenol red (21063029, Gibco) was used as the imaging solution to eliminate the short time starvation side effects during imaging process. Cells were washed with the imaging solution and imaged using an inverted microscope (Olympus IX81). The fluorescence emission excited at 488 nm and 561 nm wavelengths were acquired with an EM-CCD camera. The fluorescence intensity of pHrodo™ Green and CF555 were quantified by Image J.

RNA extraction and RT-QPCR

Total RNA was extracted using TRIzol according the manufacturer's protocol (Thermo Scientific, 191002). cDNA was generated with 100-500 ng of total RNA using GoScript Reverse Transcription System (Promega, 0000316057). Q-PCR was performed using SYBR Green (TOYOBO, 563700) in CFX Connect Optics (BIORAD). The changes in the mRNA expression of target genes were normalized to that of the housekeeping gene HPRT. Primer sequences used in this study are listed as following:

HPRT: For 5'-TGGCGTCGTGATTAGTGATG-3', Rev 5'-CTGTTCTCGTCCAGCAGACACT-3'

LAMP1: For 5'-CGTGTACGAAGGCGTTTTCAG-3', Rev 5'-CTGTTCTCGTCCAGCAGACACT-3'

ULK1: For 5'-TCATCTTCAGCCACGCTGT-3', Rev 5'-CACGGTGCTGGAACATCTC-3'

SQSTM1: For 5'-CTGGGACTGAGAAGGCTCAC-3'; Rev 5'-GCAGCTGATGGTTTGAAAT-3'

ATG5: For 5'-TGCGGTTGAGGCTCACTTTATGTC-3'; Rev 5'-GTCCCATCCAGAGCTGCTTGTC-3'

mGAPDH: For 5'-TGA ATA CGG CTA CAG CA-3'; Rev 5'-AGG CCC CTC CTG TTATTA TG-3'

mSQSTM1: For 5'-AGGAGGAGACGATGACTGGACAC-3'; Rev 5'-TTGGTCTGTAGGAGCCTGGTGAG-3'

mLC3: For 5'-CAAGCCTTCTCCTCCTGGTGAATG-3'; Rev 5'-CCATTGCTGTCCGAATGTCTCC-3'

mCTSF: 5'-ACGCCTATGCAGCCATAAAG-3'; Rev 5'-CTTTTGCCATCTGTGCTGAG-3'

Measurement of NAGase and ACP activity

Activities of NAGase and ACP enzymes were measured using microplate assay kits (NAGase, absin, abs580171; ACP, Solarbio, BC2135) respectively. Following the manufacturer's instructions, cells were seeded in 6-well plates and treated with chemicals in FBS-free medium as indicated condition. Aliquot of supernatant medium was collected and put on ice for extracellular (medium) enzyme activity detection. Cells were collected and 100 μ l of assay buffer was added. The cell suspension was then sonicated and centrifuged for 8000 $\times g$ for 10 mins, supernatant was collected for cellular enzyme activity detection. NAGase activity was measured in a 96-well microtiter plate containing 25 μ l of sample (medium or cell lysates) and 25 μ l of substrate in each well, which was mixed and incubated at 37 °C for 20 mins, and then 50 μ l of stop solution was added to stop the reaction. ACP activity was measured in a microtiter plate containing 20 μ l of sample, 40 μ l of Reagent I and 40 μ l of Reagent II. The plate was incubated at 37 °C for 15 min and then 120 μ l of Reagent III was added. Finally, the absorbance of NAGase /ACP was recorded at 405 nm/ 510 nm using a FlexStation 3 Multi-Mode microplate reader (Molecular Devices) respectively. NAGase /ACP activity released to medium (%) = Enzyme activity in medium / total enzyme activity (medium + cell lysates).

Detection of Cholesterol content

The levels of total and free cholesterol released in medium were measured using a colorimetric cholesterol/cholesteryl ester detection kit (Abcam, ab102515) according to the manufacturer's instructions. Briefly, aliquot of supernatant medium was collected and air-dried at 50 °C. The dried mixture was dissolved in cholesterol assay buffer. Aliquot of samples were mixed with cholesterol assay buffer, substrate and cholesterol enzyme with or without cholesterol esterase and incubated at 37 °C for 30 min. The absorbance was measured at 450 nm using a FlexStation 3 Multi-Mode microplate reader (Molecular Devices). The amount of cholesterol was calculated by standard curve and normalized with cellular protein content.

Cell viability assay

Cell viability was assessed by MTT assay (Merck, M2128-5G). A modified MTT assay was applied to measure the cell viability. Briefly, approximately 10^4 cells were seeded in 96-well plates and exposed to SFN as required. Cells were then incubated in fresh medium containing 0.2 mg/ml MTT solution for 4 h. The supernatant was removed and 100 μ l of acidified DMSO (0.04 M HCl/DMSO) was added to dissolve the precipitation at 37 °C for 10 min. The absorbance of dissolved solution was measured at 490 nm with Flexstation® 3 (Molecular Devices). Cell mortality (%) was calculated by $(OD_{Control} - OD_{Sample}) / OD_{Control} \times 100$.

Animals

NPC1^{-/-} mice (BALB/cNctr-Npc1 m1N/J, also known as NPC1^{NIH}) were purchased from The Jackson Laboratory (USA). All the experimental procedures were conducted in accordance with the Guide for the Care and Use of Laboratory Animals in the Zhejiang University of Technology (Hangzhou, China) and conformed to the National Institutes of Health Guide for Care and Use of Laboratory Animals. Unless stated otherwise, mice were fed with free access to water and standard diet under specific pathogen-free conditions. Genotypes were identified using a PCR-based screening (Amigo, Mendoza et al. 2002 [\[1\]](#)).

Histological analysis

Mice perfusion was performed with PBS. Then, mice cerebellums were post-fixed overnight at 4 °C and after this, placed in serial dilutions of sucrose (10-30 %) in PBS at 4°C overnight, respectively. Then cerebellums were cut in 5 µM thick sagittal sections by cryostat at (Leika) at -20 °C. Permeabilized slices with 0.1 % Triton X-100 were blocked in 1 % BSA in PBS for 1 h. Slices were incubated with anti-calbindin (Abcam, 108404) overnight at 4 °C and followed by incubation with the secondary antibody conjugated with Alexa Fluor 488 for 2 h. The slices were then washed with PBS three times and incubated with DAPI in dark for 10 min. All the images were captured by an Olympus or Zeiss confocal microscope.

Isolation of Primary MEF and macrophage cells

MEF cells were prepared from neonatal mice, which were euthanized by CO₂. Skin of neonatal mice were cut with scissors and gently clipped and added to 0.25 % trypsin-EDTA in the 37 °C incubator for 20 min. The cell suspension was then transferred to a 50 ml tube and 10 ml of DMEM media was added to inactivate the trypsin reaction for 5 min. The supernatant was transferred to a 60 mm dish and keep in incubator at 37 °C for 3 h, then the medium was replaced by DMEM with 10 % FBS, 1 % penicillin/streptomycin until confluency (2 to 4 days). Primary macrophages were established from hindlimb femurs and tibias of newborn mice. The marrow cells were flushed from the bones with PBS and centrifuged. Cells were then resuspended in RPMI 1640 medium (Gibco, B122656) supplemented with 20 % FBS. Cells were then seeded in culture dishes coated with 2 % gelatin and allowed to adhere for 2 h at 37 °C.

Reagents

Chemicals used in this study including SFN (Sigma-Aldrich, S4441), DMSO (Sigma-Aldrich, D2660), Filipin III (Cayman Chemical, 70440), U18666A (MedChemExpress, HY-107433), Triton X-100 (MCE, HY-Y1883A), NAC (Sigma-Aldrich, A7250), FK506 (Sigma-Aldrich, F4679), CsA (Solarbio, C8780), BAPTA-AM (Thermo Scientific, 1824047).

Data analysis

Data are presented as mean ± s.e.m. from at least 3 independent experiments. Statistical comparisons were performed with analyzes of variance (ANOVA) or Student's *t*-test with paired or unpaired wherever appropriate. A *p* value < 0.05 was considered as statistically significant.

Supplementary Figure Legends

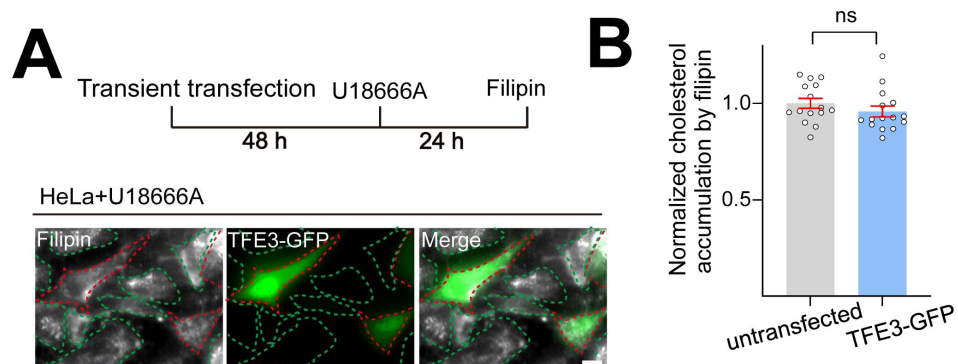


Figure S1.

The effect of TFE3 overexpression on lysosomal cholesterol levels in HeLa NPC1 model.

(A) HeLa cells were transiently transfected with TFE3-GFP plasmid for 48 h, followed by U18666A (2.5 μ M, 24 h) treatment and Filipin staining was carried out. Overlay phase contrast images are shown together with the green (TFEB-GFP) and white (filipin). In each image the red circles point to the successful transfected cells, green circles represent the untransfected cells, Scale bar, 20 μ m. **(B)** Quantification of cholesterol levels in untransfected and TFE3-GFP-transfected cells from **A**. N=15 randomly selected cells from $n=3$ independent experiments. Data are presented as mean $\pm$ s.e.m.

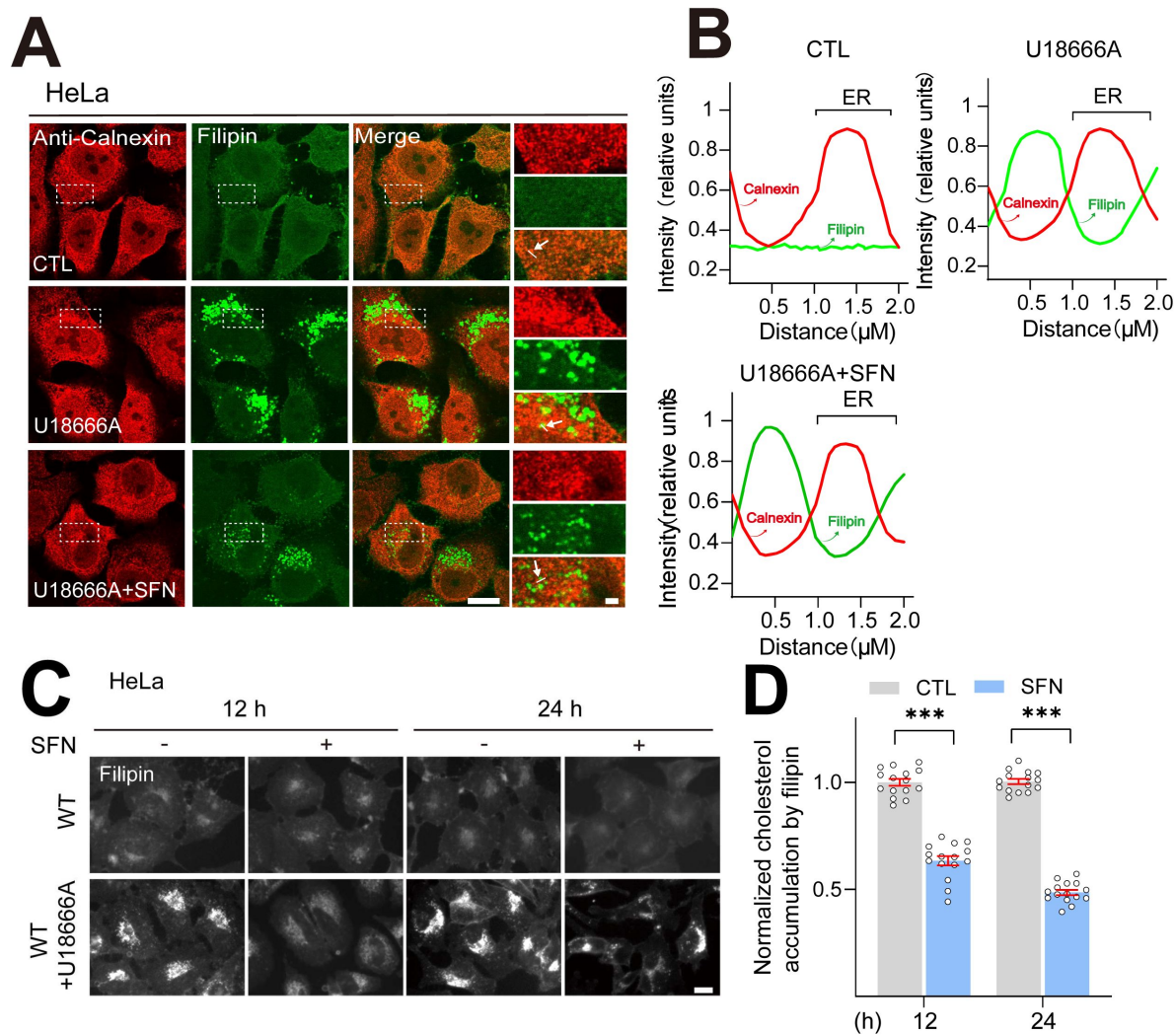


Figure S2.

Filipin signal is not colocalized with ER marker calnexin in HeLa NPC1 cells.

(A) HeLa cells were exposed to U18666A (2.5 μM) in the absence or the presence of SFN (15 μM) for 24 h. Each panel shows fluorescence images taken by confocal microscopes. The red signal is calnexin (ER marker), and the green signal is Filipin. (B) The graph shows the fluorescence intensity of a line scan (white line on the blown-up image) through the double labeled object indicated by the white arrow. (C) SFN reduces cholesterol accumulation in HeLa NPC1 cells. HeLa cells were cotreated with U18666A (2.5 μM) and SFN (15 μM) for 12-24 h and stained with filipin. Scale bar, 20 μm . (D) Quantification of cholesterol levels shown in C. N=15 randomly selected cells from $n=3$ experiments. Data are presented as mean $\pm$ s.e.m.; *** $P < 0.001$.

Figure S3.

SFN-induced TFEB nuclear translocation via a ROS-Ca²⁺-calcineurin pathway in HeLa NPC1 cells.

The effect of FK506 (5 μ M) + CsA (10 μ M), BAPTA-AM (10 μ M) or NAC (5 mM) pretreatment on SFN (15 μ M, 4 h)-mediated TFEB nuclear translocation in HeLa NPC1 cells. Scale bar: 10 μ m.
HeLa+U18666A

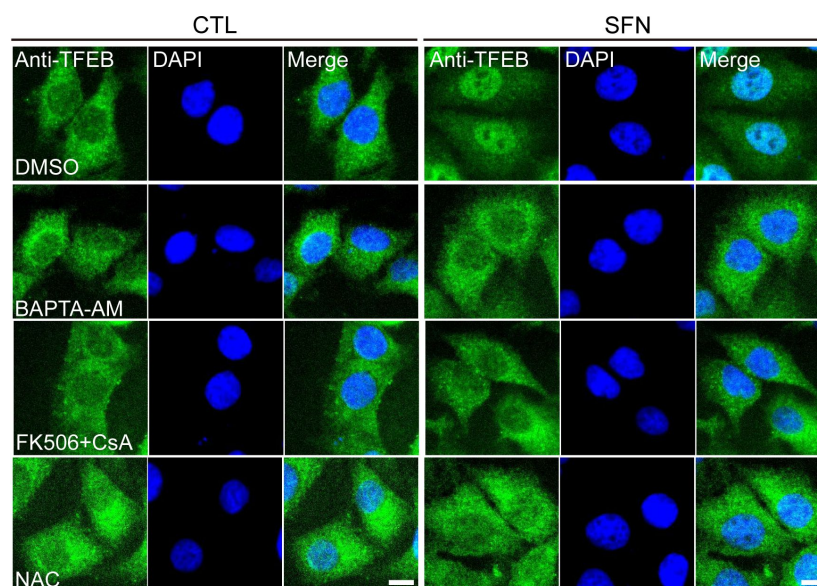


Figure S4.

Western blot analysis of the phosphorylation status of MTOR and RPS6KB1 by SFN.

(A) HeLa cells were co-treated with U18666A (2.5 μ M) and SFN (15 μ M) for 24 h. Then MTOR and RPS6KB1 activity were detected via ratios of p-MTOR vs. total MTOR and p-RPS6KB1 vs. total RPS6KB1 by Western blotting. (B) Quantification of the results shown in A. from n = 3 independent experiments. Data are presented as mean $\pm$ s.e.m..
HeLa+U18666A

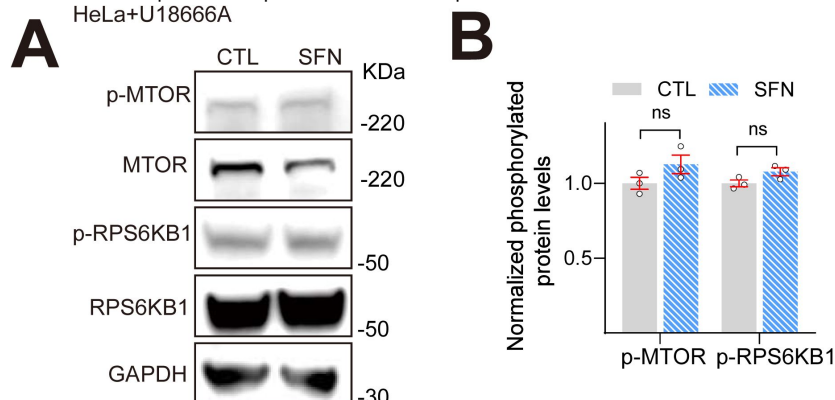


Figure S5.

Western blot analysis of NPC1 expression in human NPC1 patient fibroblasts.

Human NPC1-patient fibroblasts were treated with SFN (15 μ M) for 24 h, then the expression levels of NPC1 protein were detected by Western blotting.

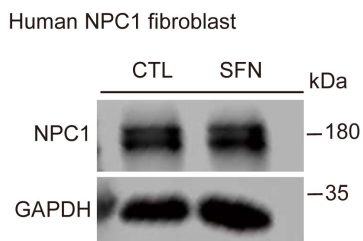
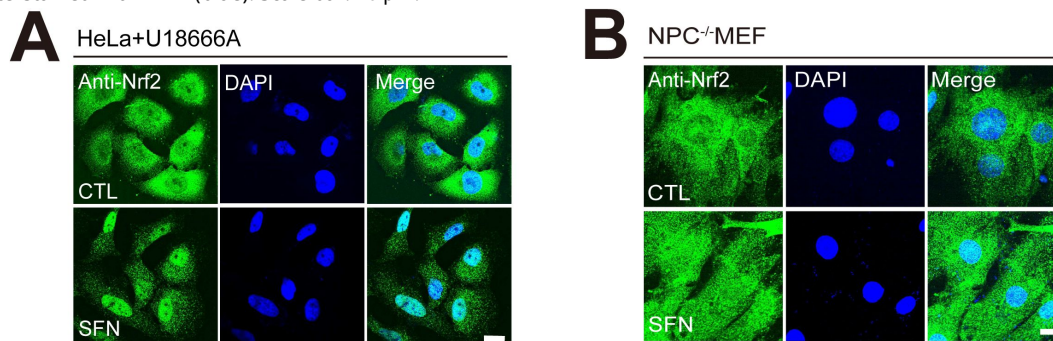


Figure S6.

SFN induced Nrf2 nuclear translocation in NPC cells.

SFN (15 μ M, 4 h) induced Nrf2 translocation from cytosol to nuclei in **(A)** HeLa NPC1 and **(B)** NPC^{-/-}MEF cells. Nuclei were counterstained with DAPI (blue). Scale bar: 10 μ m.



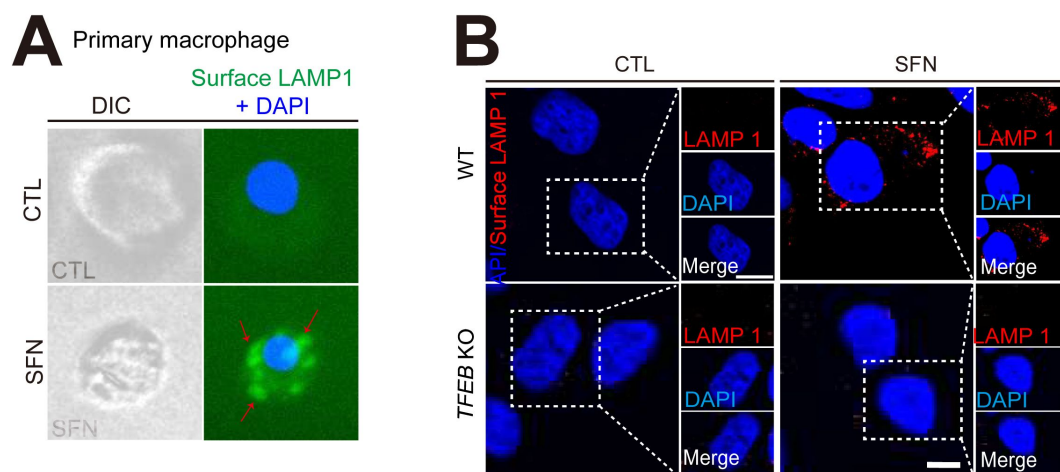


Figure S7.

SFN induces lysosomal exocytosis in a TFEB-dependent manner.

(A) Confocal microscopy images showing the increased signal of LAMP1 on the PM in nonpermeabilized primary macrophage cells with SFN (15 μ M, 12 h). Nuclei were counterstained with DAPI (blue). Scale bar, 10 μ m. **(B)** Surface LAMP1 staining in WT and *TFEB* KO HeLa cells with SFN (15 μ M, 12 h). Scale bar, 10 μ m.

Figure S8.

SFN promotes lysosomal biogenesis in *NPC1* KD HeLa cells.

(A) LAMP1 staining of *NPC1* KD HeLa cells upon SFN (15 μ M, 12 h) treatment. Nuclei were counterstained with DAPI (blue). Scale bar, 20 μ m. (B) Quantification analysis of LAMP1 immunofluorescence shown in A. N=20 randomly selected cells from $n=3$ experiments. (C) LysoTracker staining of *NPC1* KD HeLa cells with SFN (15 μ M, 12 h) treatment. Scale bar, 20 μ m. (D) Quantification of LysoTracker intensity shown in C. N=20 randomly selected cells from $n=3$ experiments. For all the panels, data are presented as mean $\pm$ s.e.m.; *** $P < 0.001$, ANOVA.

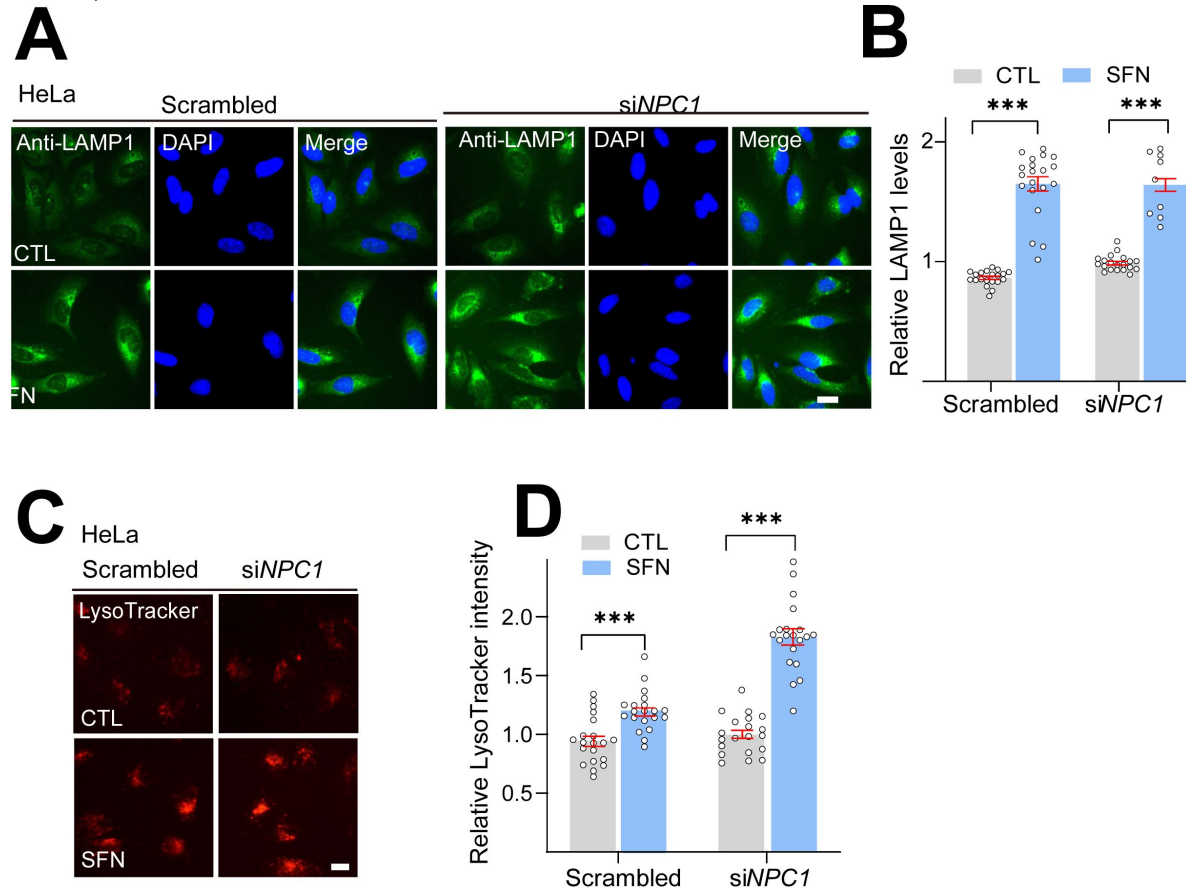
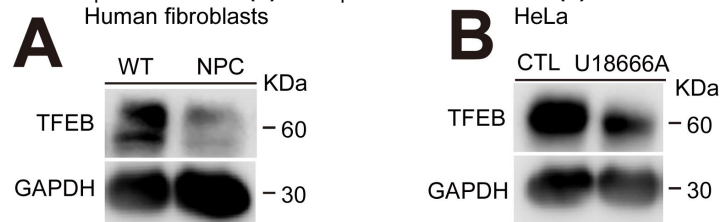


Figure S9.

TFEB expression is downregulated in *NPC1* cells.

Western blotting analysis of TFEB protein levels in (A) *NPC1*-patient fibroblasts and (B) HeLa *NPC1* cells.



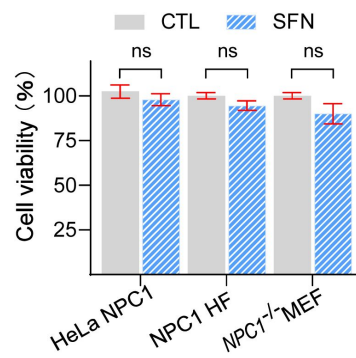


Figure S10.

The cytotoxic effects of SFN on various cell lines.

The cell viability of SFN (15 μ M, 24 h)-treated HeLa NPC1, human NPC1-patient fibroblasts, and NPC1^{-/-} MEF cells was measured by MTT assay.

Acknowledgements

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Additional information

Abbreviations

- ACP: acid phosphatase
- AREs: antioxidant response elements
- ANOVA: analyses of variance
- BBB: blood–brain barrier
- BSA: bovine serum albumin
- CLEAR: coordinated lysosomal expression and regulation
- CsA: cyclosporin A
- FBS: fetal bovine serum
- KD: knockdown
- KO: knockout
- LAMP1: lysosomal associated membrane protein 1
- ROS: reactive oxygen species
- LSDs: lysosomal storage diseases
- MEF: mouse embryonic fibroblast
- MiTF: microphthalmia/TFE transcriptional factor
- MTOR: mechanistic target of rapamycin kinase complex 1
- NAC: N-acetylcysteine
- NAGase: β -hexosaminidase
- NPC: Niemann-Pick type C disease
- OE: overexpression
- PM: plasma membrane
- RPS6 KB1/p70S6K: ribosomal protein S6 kinase B1
- SFN: sulforaphane
- TFEB: transcription factor EB

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Reviewer #2 (Public review):

Summary:

This study presents an important finding that the activation of TFEB by sulforaphane (SFN) could promote lysosomal exocytosis and biogenesis in NPC, suggesting a potential mechanism by SFN for the removal of cholesterol accumulation, which may contribute to the development of new therapeutic approaches for NPC treatment.

Strengths:

The cell-based assays are convincing, utilizing appropriate and validated methodologies to support the conclusion that SFN facilitates the removal of lysosomal cholesterol via TFEB activation.

Comments on revisions:

The authors have addressed most of my questions. I have only one minor technical point to emphasize, which does not affect the overall strength of the evidence for this project.

The pKa values of pHrodo Green (P35368, pKa=6.757) and pHrodo Red-Dex (P10361, pKa=6.816) are very similar. Prof. Xu's article, cited in the response letter (Hu, Li et al. 2022), is an excellent example of lysosomal pH measurement. He used LysoTracker Red DND-99 for a rough estimation of lysosomal acidity, and for accurate monitoring of lysosomal pH, he employed the ratiometric OG488-dex (pKa 4.6).

<https://doi.org/10.7554/eLife.103137.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Although the reviewers found our work interesting, they raised several important concerns about our study. To address these concerns, mostly we performed new experiments. The most important changes are highlighted in the summary paragraphs.

First, in response to Reviewer 1's suggestions, we have conducted the SFN experiments systematically, e.g., we further confirmed the mechanism of SFN-activated TFEB in HeLa NPC1 cells with new experiments including: the effect of BAPTA-AM (a calcium chelator), FK506+CsA (calcineurin inhibitors) and NAC (ROS scavenger) on SFN-induced TFEB-nuclear translocation in HeLa NPC1 cells (New Fig. S3). The effect of SFN on NPC1 expression (New Fig. S5). Particularly, we examined the colocalization of DiO (a PM marker) staining and surface LAMP1 staining in HeLa NPC1 cells under SFN treatment to confirm the PM exocytosis. In main text and figure legends, accuracy of sentence is thoroughly checked and defined. Hence, we have significantly improved the presentation and clarity in the revision.

Second, in response to Reviewer 2's suggestions, we have performed additional experiments to demonstrate that the role of TFEB in SFN-evoked the lysosomal exocytosis by using TFEB-KO cells (New Fig. S7B). In TFEB KO cells, this increase of surface LAMP1 signal by SFN treatment was significantly reduced, suggestive of SFN-induced exocytosis in a TFEB-dependent manner. We also investigated the effect of U18666A on CF555-dextran endocytosis. By examining the localization of CF-dex and Lamp1, we found that CF555 is present in the lysosome with U18666A treatment (Fig for reviewers only A,B), suggesting that NPC1 deficiency/U18666A treatment has no effect on CF-dex endocytosis.

Third, in response to Reviewer 3's suggestions, we have performed experiments in addition to response to other reviewers' suggestion ie. the cytotoxicity of the concentration of SFN used in this study in various cell lines (New Fig.S10).

In addition, according to the reviewers' suggestions, we made clarifications and corrections wherever appropriate in the manuscript.

Reviewer #1 (Public review):

Summary:

The authors are trying to determine if SFN treatment results in dephosphorylation of TFEB, subsequent activation of autophagy-related genes, exocytosis of lysosomes, and reduction in lysosomal cholesterol levels in models of NPC disease.

Strengths:

- (1) Clear evidence that SFN results in translocation of TFEB to the nucleus.*
- (2) In vivo data demonstrating that SFN can rescue Purkinje neuron number and weight in NPC1^{-/-} animals.*

Thank you for the support!

Weaknesses:

- (1) Lack of molecular details regarding how SFN results in dephosphorylation of TFEB leading to activation of the aforementioned pathways. Currently, datasets represent correlations.*

Thank you for raising this critical point! The reviewer is right that in this manuscript we did not talk too much about the molecular mechanism of SFN-evoked TFEB activation. Because in our previous study (Li, Shao et al. 2021), we explored the mechanism of SFN-induced TFEB activation. We show that SFN-evoked TFEB activation via a ROS-Ca²⁺-calcineurin dependent but MTOR -independent pathway (Li, Shao et al. 2021). In the current manuscript, we cited this paper, but did not talk the details of the mechanism, which obviously confused the reviewers. Therefore, in the revision manuscript we added more details of the molecular

mechanism of SFN-activated TFEB. Also, we further confirmed this mechanism in HeLa NPC1 cells with new experiments including: the effect of BAPTA-AM (a calcium chelator), FK506+CsA (calcineurin inhibitors) and NAC (ROS scavenger) on SFN-induced TFEB-nuclear translocation in NPC cells (New Fig.S3).

(2) Based on the manuscript narrative, discussion, and data it is unclear exactly how steady-state cholesterol would change in models of NPC disease following SFN treatment. Yes, there is good evidence that lysosomal flux to (and presumably across) the plasma membrane increases with SFN. However, lysosomal biogenesis genes also seem to be increasing. Given that NPC inhibition, NPC1 knockout, or NPC1 disease mutations are constitutively present and the cell models of NPC disease contain lysosomes (even with SFN) how could a simple increase in lysosomal flux decrease cholesterol levels? It would seem important to quantify the number of lysosomes per cell in each condition to begin to disentangle differences in steady state number of lysosomes, number of new lysosomes, and number of lysosomes being exocytosed.

Thank you for this constructive comment. From our data, in NPC1 cells SFN reduced the cholesterol levels by inducing lysosomal exocytosis and increasing lysosomal biogenesis. We understand the reviewer's point that it would be really helpful to differentiate the exact three states of original number of lysosomes, number of new lysosomes, and number of lysosomes being exocytosis. Unfortunately, due to the technique limitation, so far seems there is no appropriate method that could clearly differentiate the lysosomes exactly come from which state. In the future, hopefully we will have technique to explore this mechanism.

(3) Lack of evidence supporting the authors' premise that "SFN could be a good therapeutic candidate for neuropathology in NPC disease".

Suggestion was taken! We removed this sentence. Thanks!

Reviewer #2 (Public review):

(4) The in vivo experiments demonstrate the therapeutic potential of SFN for NPC. A clear dose response analysis would further strengthen the proposed therapeutic mechanism of SFN.

Thank you for this constructive suggestion. We examined the effect of two doses of SFN30 and 50mg/kg on NPC mice. As shown in Fig.6, SFN (50mg/kg), but not 30mg/kg prevents a degree of Purkinje cell loss in the lobule IV/V of cerebellum, suggesting a dose-correlated preventive effect of SFN. In the future study, we will continue optimizing the dosage form and amount of SFN and do a dose-responsive analysis.

(5) Additional data supporting the activation of TFEB by SFN for cholesterol clearance in vivo would strengthen the overall impact of the study.

Thank the reviewer for this constructive comment. We have detected a significant decrease of pS211-TFEB protein in brain tissues of NPC mice upon SFN treatment compared to vehicle, suggesting that SFN activates TFEB in brain tissue for the first time. It is worth to further examine the lysosomal cholesterol levels in brain tissues to show the direct effect of SFN. However, in our hands and in the literatures Filipin seems not suitable for detecting lysosomal cholesterol accumulation in brain tissue. So far there isn't a good method to directly measure lysosomal cholesterol in tissue.

(6) In Figure 4, the authors demonstrate increased lysosomal exocytosis and biogenesis by SFN in NPC cells. Including a TFEB-KO/KD in this assay would provide additional

| validation of whether these effects are TFEB-dependent.

Great suggestion! We investigated the role of TFEB in SFN-evoked the lysosomal exocytosis by using TFEB-KO cells. As shown in New Suppl. Fig. 7B, in TFEB KO cells, this increase of surface LAMP1 signal by SFN (15 μ M, 12 h) treatment was significantly reduced, suggestive of SFN induced exocytosis in a TFEB-dependent manner.

(7) For lysosomal pH measurement, the combination of pHrodo-dex and CF-dex enables ratiometric pH measurement. However, the pKa of pHrodo red-dex (according to Invitrogen) is ~6.8, while lysosomal pH is typically around 4.7. This discrepancy may account for the lack of observed lysosomal pH changes between WT and U18666A-treated cells. Notably, previous studies (PMID: 28742019) have reported an increase in lysosomal pH in U18666A-treated cells.

We understand the reviewer's point. But as stated in the methods and main text, we used pHrodo™ Green-Dextran (P35368, Invitrogen), rather than pHrodo Red-dextran. According to the product information from Invitrogen, pHrodo Green-dex conjugates are non-fluorescent at neural pH, but fluorescence bright green at acidic pH around 4, such as those in endosomes and lysosomes. Therefore, pHrodo Green-dex is suitable to monitor the acidity of lysosome (Hu, Li et al. 2022). We also used LysoTracker Red DND-99 (Thermo Scientific, L7528) to measure lysosomal pH (Fig. 4G, H), which is consistent with results from pHrodo Green/CF measurement.

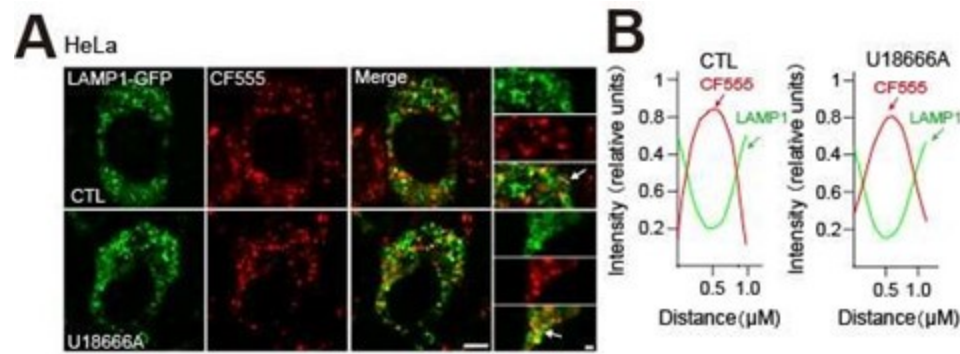
The reviewer mentioned that previous studies have reported an increase in lysosomal pH in U18666A-treated cells. We understood this concern. But in our hands, from our data with two lysosomal pH sensors, we have not detected lysosomal pH change in U18666A-treated NPC1 cell models.

(7) The authors are also encouraged to perform colocalization studies between CF-dex and a lysosomal marker, as some researchers may be concerned that NPC1 deficiency could reduce or block the trafficking of dextran along endocytosis.

Thank you for raising this important point and suggestion was taken! We investigated the effect of NPC1 deficiency on CF555-dextran trafficking into lysosome by examining the localization of CF-dex and Lamp1. To clearly define whether CF555-dex is present in the lysosome, we first used apilimod to enlarge lysosomes and then examined the relative position of CF555-dex and lamp1. As shown in Author response image 1A,B, in HeLa cells treated with U18666A, CF555 signals (red) clearly present inside lysosome (LAMP1 labelled lysosomal membrane, green signal), suggesting that CF555dex endocytosis is not affected by NPC1 deficiency (U18666A treatment).

Author response image 1.

The effect of NPC1 deficiency on CF555 endocytosis. HeLa cells were transiently transfected with LAMP1-GFP plasmid for 24 h. Cells were then treated with apilimod (100 nM) for 2 h to enlarge the lysosomes, and followed by co-treatment of U18666A (2.5 μ M, 24 h) and CF555 (12 h). (A) Each panel shows fluorescence images taken by confocal microscopes. (B) Each panel shows the fluorescence intensity of a line scan (white line) through the double labeled object indicated by the white arrow. Scale bar, 20 μ m or 2 μ m (for zoom-in images).



(9) *In vivo* data supporting the activation of TFEB by SFN for cholesterol clearance would significantly enhance the impact of the study. For example, measuring whole-animal or brain cholesterol levels would provide stronger evidence of SFN's therapeutic potential.

We really appreciate the reviewer's comments. Please see response to point #5.

Reviewer #3 (Public review):

(10) *The manuscript is extremely hard to read due to the writing; it needs careful editing for grammar and English.*

Sorry for the defects in the writing and grammar. We had thoroughly checked grammar and polished the English to improve the manuscript.

(11) *There are a number of important technical issues that need to be addressed.*

We will address the technical issues mentioned in the following ques ons.

(12) *The TFEB influence on filipin staining in Figure 1A is somewhat subtle. In the mCherry alone panels there is a transfected cell with no filipin staining and the mCherry-TFEB211A cells still show some filipin staining.*

Thank you for raising this point. The reviewer is right that not all the mCherry alone cells with the same level of filipin signal and not all mCherry-TFEB211 transfected cells show completely no filipin signal. The statistical results were from randomly selected cells from 3 independent experiments. To avoid the confusion, we have included more cells in the statistical analysis to cover all the conditions as shown in the new Fig. 1B. Hopefully this helps to clarify the confusion.

(13) *Figure 1C is impressive for the upregulation of filipin with U18666A treatment. However, SFN is used at 15 microM. This must be hitting multiple pathways. Vauzour et al (PMID: 20166144) use SFN at 10 nM to 1microM. Other manuscripts use it in the low microM range. The authors should repeat at least some key experiments using SFN at a range of concentrations from perhaps 100 nM to 5 microM. The use of 15 microM throughout is an overall concern.*

The reason that we use this concentration of SFN is based on our previous study (Li, Shao et al. 2021). We had shown that SFN (10–15 μ M, 2–9 h) induces robust TFEB nuclear translocation in a dose- and time-dependent manner in HeLa cells as well as in other human cell lines without cytotoxicity (Li, Shao et al. 2021). Also, tissue concentrations of SFN can reach 3–30 μ M upon broccoli consumption (Hu, Khor et al. 2006), so we used low micromolar

concentrations of SFN (15 μ M) in our study. Moreover, we further confirmed that SFN (15 μ M) induces TFEB nuclear translocation in HeLa NPC1 cells (Fig. 1F, G Fig. 2B, G) and this concentration of SFN has no cytotoxicity (New Fig.S10).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

The following comments are designed to improve and focus the authors' work.

(14) Related to data in Figure 1. The mechanism through which TFEB can reduce Filipin in U18 conditions is unclear. Inhibition of NPC1 results in hyperactivation of mTOR through cholesterol transport at ER-Lysosome contacts (see Zoncu group publications). If mTORC is hyperactive in NPC disease models, TFEB would be expected to remain cytoplasmic and not enter the nucleus as the representative image in Figure 1A demonstrates.

In our previous study (Li, Shao et al. 2021), we have shown that SFN induces TFEB nuclear translocation in a mTOR-independent manner (Li, Shao et al. 2021). Consistent with this result, in this study we confirmed that SFN-induced TFEB nuclear translocation is mTOR-independent in NPC1 cells (Now Fig. S4A, B). Thus, SFN induced TFEB nuclear translocation in various NPC cells (Fig. 1F, G, Fig. 2B, G). Please also see the discussion about the mechanism of SFN in response to point #1.

(15) Therefore, how does overexpression of TFEB, which remains in the cytoplasm, result in a decreased filipin signal? Similar questions relate to Figure 1C-H.

Medina et. al (Medina, Fraldi et al. 2011) show that TFEB overexpression (not activation, so overexpressed TFEB is in the cytoplasm) increases the pool of lysosomes in the proximity of the plasma membrane and promotes their fusion with PM by raising intracellular Ca^{2+} levels through lysosomal Ca^{2+} channel MCOLN1, leading to increased lysosomal exocytosis. Hence, TFEB overexpression only (TFEB is not activated) could reduce filipin signal via increasing lysosomal exocytosis. And with TFEB agonist treatment such as TFEB could further boost this increase.

(16) It would seem appropriate to measure the NPC1 and NPC2 proteins using western blot to ensure that SFN-dependent clearance of cholesterol is not due to enhanced expression of the native protein in U18-treated cells or enhanced folding of the protein in patient fibroblasts.

Thank you for this constructive comment! Because NPC1 gene mutation takes about 95% of NPC cases and NPC2 mutation takes about 5% of NPC cases. And in this study we focused on NPC1 deficiency cases. Thus, we measured the effect of SFN on the expression of NPC1 in human NPC1-patient fibroblasts. Western blot analysis showed that SFN (15 μ M, 24 h) treatment did not affect NPC1 expression in human NPC1-patient fibroblasts (new Fig. S5).

(17) Related to data in Figures 1C-E. Controls are missing related to the effect SFN has on steady-state cholesterol levels. This may be insightful in providing information on the mode of action of this compound.

Suggestion was taken! We have supplemented the control- SFN only in new Fig. 1C-E.

(18) The mechanism that links SFN to TFEB-dependent translocation is suggested to involve calcineurin independent dephosphorylation of TFEB. However, no data is provided. It would seem important to identify the mechanism(s) through which SFN positively regulates TFEB location. This would shift the manuscript and its model from correlations

to causation. Experiments involving calcineurin inhibitors, or agonists of TRPML1 that have been reported as being a key source of Ca^{2+} for calcineurin activation, may provide molecular insight.

Please see the paragraph in response to point #1.

(19) Related to Figure 4. Using a plasma membrane counterstain to quantify plasma membrane LAMP1 would increase the rigor of the analysis.

Great idea! We examined the colocalization of DiO (a PM marker) staining and LAMP1 staining in HeLa NPC1 cells under SFN treatment. As shown in new Fig.4A, surface LAMP1 signal(red) colocalized with DiO (green), a PM marker.

(20) Related to Figure 5. How do the authors explain the kinetic disparity between SFN treatment for 24 vs 72 hrs? IF TFEB is activated and promoting lysosomal biogenesis and increased lysosomal flux across the PM, why does cholesterol accumulation lag? Perhaps related to this point. Are other cholesterol metabolizing enzymes that may have altered activity in NPC sensitive to SFN? A similar comment applies to the Sterol regulatory element binding protein pathway, which has been shown to be activated in models of NPC disease.

We understand the reviewer's point. As shown in Fig. 5C, D, in NPC1^{-/-} MEF cells, SFN treatment for 24 h showed relative weaker cholesterol clearance compared to the effects in human cells (Fig.1C, D, Fig.2.E, I). Thus, we explored a longer treatment of SFN for 72 h (fresh SFN in medium was added every 24 h), and 72h treatment of SFN exhibited substantial cholesterol reduction (Fig. 5C, D). This different effect could be attributed to the continuous action of SFN, which could prolong the exocytosis, leading to more effective cholesterol clearance. As shown in the DMSO-treated MEF cells, the cholesterol levels are similar in both 24 and 72 h, thus 24 h U18666A treatment has reached the upper limit of the accumulated cholesterol, longer treatment me would not change the cholesterol levels. Thus, cholesterol accumulation has no lag.

We did not investigate whether SFN regulates other cholesterol metabolizing enzymes or sterol regulatory element binding proteins although we cannot rule out this possibility. In this study we mainly focus on the cholesterol clearance effect by SFN via TFEB-mediated pathways. From our data, TFEB KO could significantly diminish SFN-evoked cholesterol clearance. Hence, the effect of other cholesterol metabolizing enzymes or sterol regulatory element binding proteins maybe not as important as TFEB, thus out of scope of this study. In the future, we may explore the involvement of possible other pathways on SFN's effects.

(21) Related to Figure 7. The western blots for pS211-TFEB are poor. It's suggested that whole blots are shown to increase rigor.

Thank you for the comments. We have represented the blots with more spare space to increase the rigor.

(22) Data demonstrating the ability of SFN to improve Purkinje cell survival are exciting and pair well with the weight analysis, however, to address the overall goal of determining if "SFN could be a good therapeutic candidate for neuropathology in NPC disease" survival analysis should be tested as well.

Please see the paragraph in response to point #3.

Minor

(23) Throughout the manuscript many different Fonts and font sizes are used. This is very jarring to readers. It is suggested that a more uniform approach is taken to presenting these nice datasets.

We are so sorry and apologize for these oversights. We have thoroughly checked all the manuscript to make sure that Fonts and sizes of font are synchronized.

(24) Related to data presentation. In general, there is a lack of alignment and organization of the figures.

So sorry about this. We have reorganized the figures to get them better aligned.

(25) Line 149, SFN is missing.

Corrected!

Reviewer #3 (Recommendations for the authors):

(26) In Figure 3 the authors should use multiple single siRNAs or perform a functional rescue to determine specificity.

We understand the reviewer's point. We did design several siRNAs and the efficiency of these siRNAs were validated. Finally, we decide use this siRNA whose knockdown efficiency is best in the study and the specificity of the siTFEB has been validated by Western blot as shown in Fig. 3A. Furthermore, we used TFEB knockout cells constructed by CRISPR/Cas9 to further examine the role of TFEB in SFN-induced cholesterol clearance (Fig. 3D). Consistently with the results in the siTFEB-transfected HeLa NPC1 cells (Fig. 3B, C), SFN failed to diminish cholesterol in HeLa TFEB KO cells. The result from TFEB KO cells is even convincing than siRNA experiment. We also performed a functional rescue of re-expressing TFEB in TFEB KO cells, in which SFN-induced cholesterol clearance was restored (Fig. 3E, F). Collectively, these data indicate that TFEB is required for lysosomal cholesterol reduction upon SFN treatment. Thus, we did not repeat this rescue experiment in the siTFEB-transfected HeLa NPC1 cells.

(27) The label for 3D is missing.

Corrected! Thanks!

(28) Figure 4, although the authors use an antibody against the luminal domain of LAMP1 there could still be some permeabilization. A marker of the plasma membrane would be helpful.

Please see the response to point #19.

(29) Figure 4, cholesterol in the media because of lysosome exocytosis. This is where the high concentration of SFN is of concern. Is there any cell death that could explain the result? The authors should test for cell death with the SFN treatment.

Thank you for raising this important point! We have measured the cytotoxicity of SFN of the concentrations used in this study in various cell lines (New Fig.S10). Please also see the paragraph in response to point #13.

(30) The blot in Figure 6A is unclear. It is very hard to see any change in pS211-TFEB levels, and, the blurry signal is the detection of phospho-TFEB is uncertain.

Please see the summary paragraph in response to point #21.

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